Until we are all equal

Participatory Climate Risk and Vulnerability Assessment for Gender Responsive Climate Adaptation and Preparedness Plans (GRAPP) in Kwale.

Draft Kwale County Climate Change Adaptation Action Plan (2025–2030)

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1 Chapter 1: Background information

1.1 Introduction

Plan International Kenya is implementing the Conservation and Sustainable Management of Marine and Coastal Ecosystems (COSME) Project in Kwale and Kilifi Counties, two coastal counties endowed with ecologically and economically significant marine and terrestrial ecosystems. These include mangrove forests, coral reefs, seagrass beds, coastal forests, and extensive open-sea marine resources that support key livelihood sectors such as tourism, fisheries, agriculture, and maritime trade. These ecosystems provide essential services including shoreline protection, carbon sequestration, biodiversity conservation, and nutrient cycling forming the ecological foundation of the counties' economies.

Kwale County falls within a region of high climate sensitivity. The county covers approximately 8,270 km² with a population of 866,820 persons (426,266 males and 440,554 females) (KNBS, 2019). The climate is predominantly tropical, with bimodal rainfall (March–May and October–December) and annual temperatures ranging between 24°C and 30°C. Kwale's economy is primarily driven by climate-dependent sectors; rain-fed agriculture, livestock rearing, fisheries, tourism, and mining, making it highly vulnerable to climatic shocks and stresses. According to the 2021 County Climate Risk Profile, the county faces recurrent hazards including droughts, floods, sea level rise, ocean acidification, saltwater intrusion, coastal erosion, and forest degradation. These climatic risks are already impacting livelihoods, infrastructure, and ecosystems, resulting in declining productivity, increased poverty levels, and heightened vulnerability among marginalized groups such as women, youth, and Persons Abled Differently (PAD).

To respond to these challenges, the COSME project adopts an ecosystem-based and gender-responsive approach to strengthen community resilience and promote adaptive livelihoods. The initiative supports the formulation of Gender Responsive Adaptation and Preparedness Plans (GRAPPs) which combine participatory climate risk assessments with evidence-based adaptation planning. These ward-level plans form the foundation for the development of the Kwale County Climate Change Adaptation Action Plan (CCAAP 2025–2030), ensuring that county-level strategies are anchored in local realities and inclusive decision-making processes.

As part of the COSME project, three wards; Pongwe Kikoneni, Ramisi, and Vanga, were sampled to represent the county's coastal ecological and socio-economic contexts. These wards were selected based on their exposure to coastal and marine climate risks, dependence on natural resources, and the presence of vulnerable populations.

These localized vulnerabilities underscore the urgency of adaptive action at both the ward and county levels. Accordingly, the Kwale County Climate Change Adaptation Action Plan (CCAAP) consolidates findings from the ward adaptation processes, aligning them with national frameworks such as the Climate Change Act (2023), National Climate Change Action Plan (NCCAP), and National Adaptation Plan (NAP). This alignment ensures consistency with Kenya's devolved climate governance agenda while enhancing local ownership, equity, and gender inclusivity in adaptation planning and implementation.

This plan aims to mainstream climate resilience into all sectors, safeguard ecosystem integrity and build adaptive capacities of communities, particularly women and other vulnerable groups to respond effectively to the increasing impacts of climate change.

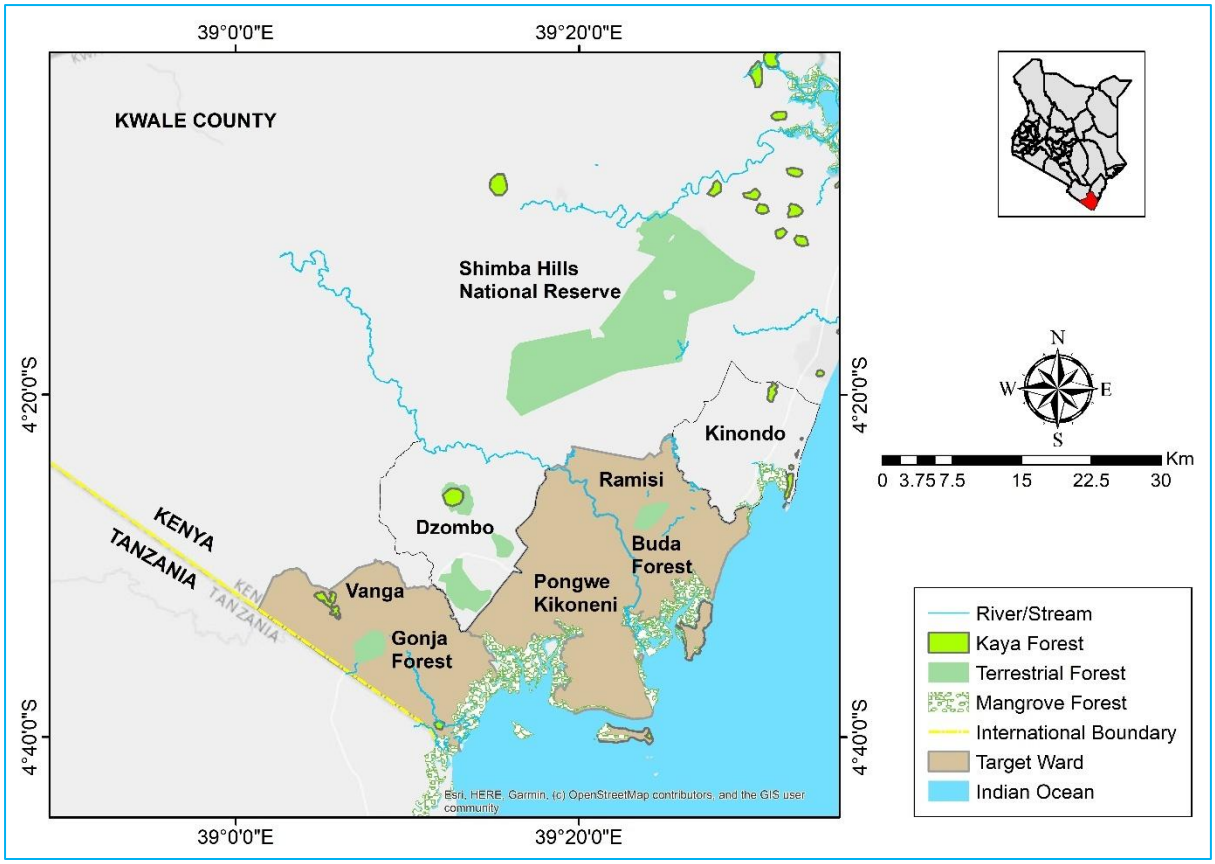

Figure 1: Location of Kwale County and survey sites (Source: PCVA, 2025).

1.2 Plan development approach

The Kwale County Climate Change Adaptation Action Plan (CCAAP 2025–2030) was developed through a participatory, evidence-based, and gender-responsive process that builds upon the foundations laid by the Ward Adaptation Plans (WAPs) for Pongwe Kikoneni, Ramisi, and Vanga Wards. The plan draws extensively from Participatory Climate Vulnerability and Risk Assessments (PCVAs) conducted across fourteen (14) villages, namely Anzuwani, Kiwambale, Barabarani, Mkwiro, and Kibuyuni in Pongwe Kikoneni Ward; Mafisini, Dzibwage, Darigube, Nyumba Sita, and Mwangundu in Ramisi Ward and Mwalewa C, Sega D, Ngweneni, and Mikoroshoni in Vanga Ward. These sites were selected to represent the ecological, socio-economic, and cultural diversity of Kwale County's coastal landscape, while also reflecting areas of high climate exposure and ecological sensitivity within the COSME Project's intervention scope.

The development of the CCAAP was anchored in a livelihoods-based vulnerability framework that recognizes vulnerability as the outcome of interrelated factors i.e., exposure, sensitivity, and adaptive capacity, shaped by ecological conditions, social networks, economic assets, and institutional systems (Figure 2). This framework underscores that vulnerability is experienced differently across population groups, with women, men, youth, and Persons Abled Differently (PAD) facing unique climate risks and adaptation barriers. By adopting this gender-responsive perspective, the CCAAP ensures that the identified adaptation measures address the differentiated needs, capacities, and priorities of diverse social groups within the county.

The CCAP was therefore designed using a process that combined experiential community knowledge with quantitative datasets, spatial analysis, ecological diagnostics and climatic modelling. This integrated methodology enabled the plan to link climatic hazards to specific livelihood systems; rainfed agriculture, artisanal fishing, seaweed farming, mangrove harvesting,

small-scale trade and casual labour and to examine how these systems are shaped by gender norms, social vulnerabilities and institutional arrangements at ward level. The resulting insights provided a nuanced understanding of how climate extremes intersect with poverty, food insecurity, ecosystem degradation, weak infrastructure and limited institutional support, thereby shaping the context within which adaptation strategies must be developed.

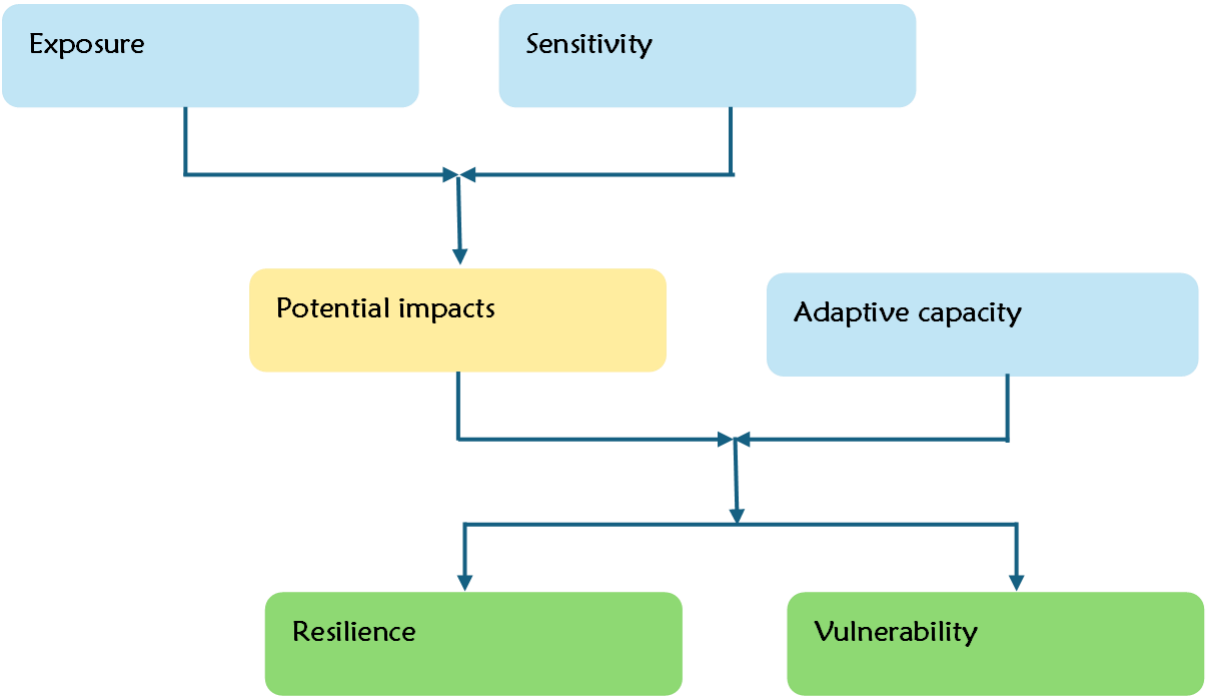

Figure 2: A model of factors that can be used to assess the vulnerability and resilience of systems (Source: Füssel and Klein, 2006, and Gauthier et al., 2014).

1.2.1 Inception Phase

The methodology was agreed upon through an inception phase that established a shared understanding among Plan International Kenya and the consultant on the objectives, scope and analytical expectations of the PCVA and WAP processes. An inception meeting held on 5th September 2025 brought together the COSME project teams to refine the methodological sequence, expected results, deliverable structure and timelines. During this phase, existing climate risk information for Kwale County was reviewed, data gaps specific to the three wards were identified, and a sequenced work plan was developed outlining participatory diagnostics, household surveys, spatial and climate analysis, and subsequent synthesis and validation activities.

Following institutional alignment, data collection instruments were developed and technically refined to ensure that they were methodologically sound, gender-responsive and culturally appropriate. The instruments included ward-level FGD guides segregated for women, men and youth; participatory mapping and historical timeline guides; seasonal calendar and daily clock templates; household decision-making pile-sorting tools; Leaky Bucket diagrams; impact chain protocols; vulnerability matrix templates; institutional Venn diagram formats; and adaptation pathway worksheets. Each tool integrated prompts on gender norms, intersectional vulnerability, unpaid care workloads, labour shifts during climate stress and disability inclusion.

Stakeholder mapping during the inception phase identified relevant actors within the county, including the County Climate Change Planning Committee, ward administrators, chiefs and assistant chiefs, village elders, leaders of women and youth groups, PAD representatives and support networks, seaweed farming groups, mangrove user groups, BMUs, WRUAs and community-based organizations. These actors facilitated community mobilization, venue preparation and dissemination of study information at ward level. The inception phase ensured that ward stakeholders were adequately oriented and positioned to meaningfully participate in the PCVA and adaptation planning process.

1.2.2 Participatory Climate Risk and Vulnerability Assessment (PCVA)

The CCAAP was informed by a comprehensive PCVA process implemented in 14 COSME target villages within Pongwe Kikoneni, Ramisi and Vanga wards. The assessment followed a structured six-step participatory sequence designed to diagnose climate vulnerabilities, identify underlying drivers of sensitivity, assess existing coping and adaptation strategies, and translate diagnostic findings into actionable adaptation options suitable for inclusion in the CCAAP. The process applied the full suite of ten PCVA tools, each deployed to generate gender- and age-disaggregated evidence capable of capturing complex interactions between climate hazards, livelihoods, natural resources and social systems.

1.2.2.1 Step 1: Scoping

During scoping, the geographical scope of the county was delineated using GIS shapefiles provided by Plan International and Kwale County Government. Within this boundary, the 14 COSME villages were purposively selected on the basis of ecological sensitivity, concentration of climate-sensitive livelihoods, recurrent exposure to droughts and floods, susceptibility to strong winds and sea-water intrusion, and documented human–wildlife conflict incidents.Village profiles were then developed to summarize demographic characteristics, livelihood systems, existing infrastructure, ecosystem conditions, climate-related impacts, past interventions and vulnerability hotspots. These profiles served as the baseline data against which PCVA tools were deployed.

1.2.2.2 Step 2: Establishment of Core Participatory Analysis Teams

Ward-level core participatory analysis teams were established for Pongwe Kikoneni, Ramisi and Vanga drawing membership from Ward Climate Change Planning Committee (WCCPC) representatives, village elders, women leaders, youth leaders, PAD representatives and carers, NbS groups, GJJ groups, and other active CBOs and self-help groups. County technical officers from climate change, environment, forestry, agriculture, livestock and fisheries were engaged as resource persons.

The core team underwent a structured orientation on the PCVA framework, the ten PCVA tools, gender- and age-responsive facilitation, child safeguarding and ethical data collection. Training sessions used a combination of plenary presentations, group work and simulations of each PCVA tool, conducted primarily in Kiswahili to ensure comprehension. By the end of the training, the ward team had the capacity to co-facilitate hazard and resource mapping, historical timelines, seasonal calendars, daily clocks, household decision making and pile-sorting exercises, Leaky Bucket analyses, impact chain mapping, vulnerability matrix scoring, Venn diagrams and adaptation pathway development.

1.2.2.3 Step 3: Situation Analysis of Climate Vulnerability

The situation analysis integrated secondary datasets, primary field evidence and participatory analytics to generate a comprehensive picture of exposure, sensitivity and adaptive capacity across the three wards. This phase synthesised literature review, ward-level gender- and age-disaggregated Focus Group Discussions (FGDs), transect walks across the 14 COSME villages,

and the structured application of PCVA Tools 1–7. Through triangulation, these methods produced a detailed, spatially nuanced and socially differentiated understanding of how climatic hazards interact with biophysical systems, livelihood structures and gendered social dynamics.

1.2.2.3.1 Literature Review

The literature review served as the technical backbone of the situation analysis, drawing from global, regional, national, county and historical ward-level evidence to contextualize observed vulnerabilities. At the global scale, the review considered findings from IPCC AR6 Working Group I and II, particularly projections relevant to East African coastal systems, including accelerated warming rates, sea surface temperature rise, intensification of short-duration extreme rainfall, increased occurrence of coastal storms, saline intrusion risks and anticipated sea-level rise along the Western Indian Ocean. These findings were aligned with global adaptation and risk governance frameworks such as the Sendai Framework for Disaster Risk Reduction 2015–2030, the Paris Agreement adaptation cycle, and SDGs 2, 5, 6, 13 and 14, which collectively frame resilience building in coastal and marine socio-ecological systems.

At the regional scale, the review integrated IGAD Climate Prediction and Applications Centre (ICPAC) bulletins, the Africa Climate Policy Centre (ACPC) projections for Eastern Africa, and Western Indian Ocean Marine Science Association (WIOMSA) research on mangrove decline, seaweed vulnerability, fishery collapses and oceanographic shifts. This regional evidence positioned Ramisi within a broader coastal climate risk corridor characterized by rising temperatures, altered monsoon circulation, changing upwelling systems and degradation of nearshore ecosystems.

National climate policy and regulatory frameworks were examined comprehensively, including the Climate Change Act (2023), National Adaptation Plan (NAP 2015–2030), Kenya's Updated NDC (2023), the National Framework for Climate Services (NFCS), Kenya Climate-Smart Agriculture Strategy, National Disaster Management Policy, National Drought Early Warning System, and EMCA (2015) and sectoral regulations. These documents provided the lens for interpreting institutional mandates, priority hazards and adaptation responsibilities that cascade to county and ward levels.

At county scale, the review consolidated information from the Kwale County Integrated Development Plan (CIDP III, 2023–2027), the Kwale County Climate Change Act (2022), the County Climate Change Fund (CCCF) Regulations, the County Climate Change Action Plan (CCCAP 2023–2027), the County Spatial Plan, and sector strategies in agriculture, livestock, fisheries, water and coastal management. These county frameworks were triangulated with socio-economic and food security trends reported by the National Drought Management Authority (NDMA), historical rainfall data from the Kenya Meteorological Department (KMD), marine ecosystem studies from KMFRI, forestry and mangrove assessments by the Kenya Forest Service (KFS), fishery catch records from the State Department for Fisheries, and demographic data from the Kenya Population and Housing Census (2019).

The review also incorporated previous PCVAs conducted under COSME and other coastal adaptation programmes, which provided comparative vulnerability patterns, gender-responsive insights, and past lessons on nature-based solutions, community-based early warning systems, and local adaptation practices. This multilevel synthesis created a scientifically grounded baseline against which primary data from the wards could be interpreted.

1.2.2.3.2 Focus Group Discussions (FGDs)

Ward-level FGDs were conducted with segregated groups of women, men and youth from the 14 COSME villages to elicit gendered and generational insights into climate trends, livelihood impacts and socio-ecological vulnerability (Figure 3). The discussions explored perceived changes in rainfall reliability, including delayed onset and inconsistent cessation of rains, prolonged intra-seasonal dry spells, unprecedented periods of elevated temperatures, shifts in wind patterns and increased intensity of episodic storms and coastal surges. Participants described the material impacts of these climatic variations on agricultural productivity, seaweed quality and harvest cycles, artisanal fisheries (including reduced catch per unit effort), mangrove health, livestock mortality and food security.

The FGDs also highlighted the deeply gendered nature of climate vulnerability. Women described increased walking distances and time burdens for water and fuel collection during droughts; heightened caregiving demands during disease outbreaks; increased exposure to GBV during water scarcity periods; and disruptions in women's informal trading activities. Youth reported loss of wage labour opportunities during climate shocks, stress-related migration, and lack of access to climate-resilient livelihood assets. Men described reduced fishing days caused by strong winds, damaged boats during storms and increased anxiety over household provisioning responsibilities. In addition, participants highlighted PAD-specific vulnerabilities, including mobility limitations during floods and exclusion from labour-intensive coping strategies.

These discussions produced climate calendars, gender-disaggregated vulnerability matrices and hazard trend narratives that deepened the temporal and social dimensionality of the vulnerability assessment.

Figure 3: FGD in Ramisi Ward (Source: Field Surveys, 2025)

1.2.2.3.3 Transect Walks

Transect walks undertaken across the 14 villages provided on-ground spatial verification of vulnerabilities identified in FGDs and documented in literature. These walks traversed rainfed agricultural plots, irrigated horticultural fields, mangrove stands and adjacent intertidal zones, seaweed farming lines, fish landing sites, riparian zones of River Ramisi and Umba, boreholes and shallow wells, public institutions (schools, health dispensaries), access roads, market areas, and known flood plains.

Structured observation sheets and GPS-enabled devices were used to capture geospatial attributes of key vulnerability hotspots. Evidence recorded included deterioration of feeder roads due to repeated flooding, heavily eroded gullies formed by runoff from bare catchments, saline intrusion into agricultural fields and domestic wells, siltation of wetlands, collapsed seaweed lines following strong winds, degraded mangrove stands exhibiting canopy thinning or sediment stress, and infrastructure damage such as washed-out culverts and submerged pathways.

These observations enhanced the physical realism of the vulnerability analysis and provided geo-referenced evidence that later strengthened hazard–resource maps and the identification of priority zones for ecological restoration and climate-resilient infrastructure.

1.2.2.3.4 Tool 1: Hazard and Resource Mapping

The hazard and resource mapping tool enabled community members to visually represent their environmental and socio-economic landscape. Through participatory sketching, communities identified settlement clusters, farmlands, grazing areas, mangrove forests, seaweed farms, rivers, swamps, markets, schools, health facilities, water points and religious centres. Hazard features; such as recurrent flood zones, storm-surge-prone coastal segments, areas with saline intrusion, strong-wind corridors and crocodile-risk stretches along River Ramisi and Umba, were marked and discussed. This mapping exercise generated spatially explicit knowledge that later informed adaptation targeting and zoning.

1.2.2.3.5 Tool 2: Historical Timelines

Historical timelines reconstructed multi-decadal climatic and socio-economic events including major droughts, destructive floods, storm events, pest invasions (armyworms, whiteflies and rice pests), fishery fluctuations linked to monsoon variability, severe seaweed farm losses from temperature anomalies and episodes of heightened human–wildlife conflict. Each event was discussed in terms of magnitude, social and economic impact, community responses and institutional support. Timelines revealed a pattern of intensifying climatic shocks, shortened recovery cycles and cumulative stress on natural resource systems.

1.2.2.3.6 Tool 3: Seasonal Calendars

Seasonal calendars detailed intra-annual cycles of rainfall, temperature, wind regimes, tidal patterns, agricultural operations, fishing seasons, seaweed planting and harvesting, mangrove harvesting, disease outbreaks and social events. Comparisons between historical and current seasonal behaviour revealed declining predictability of rainfall, increased occurrence of off-season rainfall, persistent dry spells and shifts in traditional fishing and seaweed farming seasons. Calendars also captured periods of heightened GBV, school absenteeism, food scarcity and increased care burdens.

1.2.2.3.7 Tool 4: Daily Clock

The daily clock exercise revealed gendered labour allocation and how climate stress reconfigures time use. Under climate-normal conditions, tasks such as farming, seaweed farming, childcare, water collection and marketing followed relatively stable patterns. Under climate stress, women

and girls reported significant increases in unpaid care work and water collection time, men noted prolonged periods of income loss due to weather-inhibited fishing, and youth described increased domestic workload substitution and reduced learning hours. These insights demonstrated how climate shocks deepen time-poverty and reduce adaptive capacity.

1.2.2.3.8 Tool 5: Household Decision-Making Pile Sorting

Pile sorting exercises clarified intra-household power/authority structures under normal and climate-stress scenarios. Under normal conditions, decision-making on crop choices, livestock sales and minor household investments tended to be joint or gender-specific depending on cultural norms. Under climate stress, however, the decision space often shifted toward male dominance, particularly in decisions involving financial risk, fishing activity, asset disposal or climate-response investment, which in turn constrained women's adaptive agency and reinforced gender-based vulnerabilities.

1.2.2.3.9 Tool 6: Leaky Bucket

The Leaky Bucket tool illustrated how climate shocks disrupt household financial stability. Communities mapped income streams such as farming, fishing, seaweed sales, petty trade, remittances and casual labour, alongside expenditure flows including food purchases, health costs, water, school fees, debt servicing and transport. Climate shocks were shown to reduce inflows drastically while simultaneously raising outflows through rising food prices, repair costs and medical expenses. This widening income–expenditure gap demonstrated the structural economic vulnerability of households.

1.2.2.3.10 Tool 7: Impact Chains

Impact chain analysis depicted the cascading consequences (direct and indirect impacts) of priority hazards. For instance, prolonged drought triggered crop losses, water scarcity, livestock mortality and reduced milk intake, which in turn contributed to household food insecurity, malnutrition risk among children, increased labour migration by men, rising care burdens for women, heightened GBV risk and reduced school attendance. Similarly, strong winds disrupted fishing and seaweed farming, leading to economic losses, debt accumulation and psychological stress. This tool exposed multi-dimensional vulnerability pathways and supported the design of integrated adaptation strategies.

1.2.2.4 Step 4: Analysis of Causes of Vulnerability (Tool 8 – Vulnerability Matrix)

The analysis of underlying causes of vulnerability was undertaken through the application of the vulnerability matrix (Tool 8), which provided a structured means of examining how different climatic hazards intersect with livelihood assets and socio-ecological systems in the three wards. Hazards such as drought, floods, environmental degradation, strong winds, sea-level rise, human–wildlife conflict and agricultural pests and diseases were arranged against critical assets including rain-fed farms, irrigated plots, mangrove forests, seaweed farms, artisanal fisheries, livestock, domestic water sources, housing structures, schools, health facilities, feeder roads and market infrastructure. Community members assigned severity ratings for each hazard–asset interaction, supported by qualitative reasoning and lived examples.

The cumulative scoring process produced a ranking that demonstrated, for example, the high susceptibility of mangrove areas and seaweed farms to combined effects of temperature stress, strong winds and changing salinity regimes; the acute vulnerability of rain-fed agriculture to prolonged drought and unpredictable rainfall; and the differentiated impacts of floods on schools and feeder roads which disrupt mobility, market access and educational continuity. This step bridged descriptive hazard analysis with deeper socio-economic and institutional determinants of vulnerability, forming the foundation for identifying priority adaptation pathways.

1.2.2.5 Step 5: Analysis of Community Action (Tool 9 – Venn Diagrams)

Analysis of existing coping and adaptation practices revealed a range of coping strategies and externally supported interventions. The Venn diagram exercise (Tool 9) enabled communities to map the institutional ecosystem influencing adaptation in the ward. Institutions such as county line departments, NGOs, community-based organisations, religious institutions, savings and credit groups, fisheries and forestry officers, COSME project structures, and private sector actors were placed on diagrams according to their perceived influence and accessibility. The spatial arrangement revealed a significant presence of institutions but varying degrees of functional support. While some organisations maintained close, supportive relationships, such as community groups collaborating with NbS and seaweed farmer associations, others operated at a distance, limiting their impact. This analysis also identified critical gaps, including insufficient extension services for seaweed and fisheries, weak linkages to financial services, limited climate information dissemination, and inadequate institutional frameworks for systematic PAD inclusion. These findings provided essential insights for designing governance and institutional strengthening elements within the County Adaptation Plan.

1.2.2.6 Step 6: Drawing Action from Analysis (Tool 10 – Adaptation Pathways)

The final step in the participatory diagnostic process involved translating identified vulnerabilities, exposure pathways and institutional gaps into a coherent adaptation strategy through the Adaptation Pathways tool (Tool 10). Communities revisited impact chains and vulnerability dynamics to identify feasible response strategies across short-, medium- and long-term horizons. The exercise involved structured deliberations that considered gendered labour implications, the availability of financial and natural resources, household and community capacities, and alignment with ongoing COSME and county initiatives.

Resulting adaptation pathways articulated sequenced actions for key socio-ecological systems; agriculture, water resources, mangrove and terrestrial forests, seaweed farming and fisheries, community infrastructure, early warning systems and social protection mechanisms. This included immediate resilience-building measures such as water harvesting, reinforcement of seaweed lines, community drainage clearance and household-level livelihood diversification; medium-term actions such as improved coastal protection, establishment of climate-resilient seaweed strains, expansion of irrigation infrastructure and strengthening local governance structures; and long-term interventions such as large-scale mangrove rehabilitation, institutionalization of community-based climate monitoring systems, and improvement of rural infrastructure. These pathways formed the backbone of the County Adaptation Plan, linking diagnostic insights to strategic action.

1.2.3 Household Survey

A structured household survey was administered to 295 households in the study area to provide quantitative evidence complementing qualitative findings. Sampling was designed to include households across diverse livelihood portfolios and different vulnerability categories, including female-headed households, large households with high dependency ratios, households with PAD members, and those primarily reliant on climate-sensitive livelihoods such as rain-fed agriculture, artisanal fishing and seaweed farming. Primary respondents were predominantly women (approximately 65 %), ensuring robust capture of intra-household dynamics, care burdens, access to services and lived experiences of climate stress.

The survey collected data on demographic structure, education, disability status, land tenure and resource access, livelihood contributions, income diversification, exposure to recent climatic shocks, losses and damages, coping strategies, food security indicators, institutional linkages and participation in social protection and savings groups. Data were collected digitally using KoboToolbox to ensure consistency and reduce transcription errors. These data provided a

statistically grounded basis for validating PCVA findings and understanding patterns of vulnerability at household level across gender, age and livelihood categories.

1.2.4 Current Mangrove and Forest Health Assessment

To assess the condition of key coastal and terrestrial ecosystems supporting nature-based adaptation in the county, a remote sensing analysis was conducted using Sentinel-2 imagery at 10-metre spatial resolution. Imagery underwent standard pre-processing steps including radiometric correction and cloud masking before the Normalized Difference Vegetation Index (NDVI) was computed using the formula.

$$\text{NDVI} = \frac{NIR - RED}{NIR + RED} \qquad \text{Equation 1}$$

Where RED is visible red reflectance, and NIR is near infrared reflectance.

NDVI values were categorized into low, moderate and high vegetation vigor classes. Low NDVI signatures corresponded to degraded or stressed vegetation, often reflecting factors such as saltwater intrusion, wood extraction pressure, sedimentation and hydrological alteration. High NDVI zones served as ecological benchmarks, indicating healthier vegetation stands. These spatial patterns were subsequently compared with PCVA hazard–resource maps and field observations to identify priority sites for ecological restoration, mangrove replanting, mangrove hydrological rehabilitation and sustainable forest management within the ward.

1.2.5 Current and Future Climate Projections

Climate projections relevant to the county were developed supplemented by remote sensing and historical observational datasets. Daily rainfall Climate Hazards Group InfraRed Precipitation with Station Data (CHIRPS), Actual Evapotranspiration (AET), Potential Evapotranspiration (PET), (from ERA5-Ag) and temperature variables (minimum, mean, maximum and Land Surface Temperature (LST) from Landsat and ERA5-Ag) were extracted through Google Earth Engine after spatial boundaries had been validated in QGIS. The resulting datasets were aggregated into annual totals or means and subsequently analysed using Python (Colab) employing pandas, numpy, scikit-learn and matplotlib.

Linear regression models were fitted to each climatic variable to estimate historical trends and project future changes to 2035 and 2045. Temperature projections incorporated Intergovernmental Panel on Climate Change (IPCC) AR6-aligned warming increments for East Africa's coastal belt, applying approximately +0.02°C per year for the moderate-emission scenario and +0.035°C per year for high-emission trajectories. Rainfall projections followed empirically derived trends, showing increasing variability, changes in seasonal distribution and heightened likelihood of intense episodic rainfall events. These climatic insights were interpreted in relation to Kwale's hazard profile, reinforcing the need for enhanced water security, climate-resilient agricultural practices, improved coastal and marine ecosystem management, stronger infrastructure planning and expanded early warning systems.

1.2.6 Data Analysis and Reporting

Quantitative data from the household surveys were cleaned, coded and analyzed using Microsoft Excel and Python. Analysis focused on frequencies, proportions, cross-tabulations and simple correlations across gender, age group, livelihood type and vulnerability status. These results provided empirical confirmation of patterns emerging from qualitative insights.

Qualitative data from WAPs, FGDs, PCVA exercises and transect walks were transcribed and thematically coded to organise evidence under key domains such as hazard exposure, livelihood impacts, gender dynamics, coping strategies, institutional barriers and opportunities. Outputs from PCVA tools; maps, timelines, calendars, clocks, matrices and impact chains, were digitized and integrated with NDVI layers and climate trend visualizations to produce synthesized analytical outputs. Triangulation across all sources ensured internal consistency and improved the robustness of the final interpretation. Draft findings were compiled for review by Plan International Kenya in line with expected deliverables.

1.2.7 Validation

The draft findings were presented in county-level validation workshops held Kwale on 21st November 2025. The workshop provided a platform for county technical departments, CCCPC/WCCPC members, civil society, COSME partners and community representatives to review the PCVA results and refine the draft adaptation strategies. At ward level, stakeholders from the three wards reviewed hazard rankings, adaptation priorities and institutional roles, and provided further refinement on feasibility, gender inclusion, sequencing and resource implications. Inputs from these processes were incorporated into the final Ward Adaptation Plan to ensure scientific accuracy, community legitimacy and institutional alignment.

1.3 Linkages to other planning processes

The Kwale County Climate Change Adaptation Action Plan (CCAAP 2025–2030) builds on existing national, county, and ward-level planning frameworks to enhance coherence, coordination, and integration of climate adaptation across governance levels. The plan draws directly from the Participatory Climate Vulnerability and Risk Assessments (PCVAs) conducted in fourteen villages across Pongwe Kikoneni, Ramisi, and Vanga Wards, and from the resulting Ward Adaptation Plans (WAPs) developed under the COSME Project, which provided detailed understanding of climate risks, vulnerabilities, and adaptive capacities within coastal communities. At the national level, the plan aligns with the National Adaptation Plan (NAP 2015–2030), which operationalizes the National Climate Change Action Plan (NCCAP) by embedding adaptation across all sectors of development through a multi-sectoral and multi-level governance framework that recognizes climate change as a cross-cutting issue influencing Kenya's economic growth, social well-being, and environmental stability. At the county level, the CCAAP complements the Kwale County Climate Change Adaptation Plan (CCAP 2023–2032), which localizes the NAP's priorities within Kwale's ecological and socio-economic context and integrates adaptation objectives into the County Integrated Development Plan (CIDP) and Annual Development Plans (ADPs). CCAAP translates these strategic directions into locally driven, gender-responsive adaptation measures that address community priorities identified through participatory assessments. By linking ward-level findings to county and national frameworks, the plan promotes vertical integration of climate action, ensuring that localized initiatives contribute to broader resilience and sustainable development outcomes across Kwale County.

2 Chapter 2: Coordination of Adaptation Actions

The Climate Change Adaptation Action Plan is guided by a multi-level governance framework spanning national, county, ward, and community institutions. At the national level, Kenya's policy and legal framework establishes the mandates and coordination structures for climate action. The Climate Change Act (2023) and the National Adaptation Plan (NAP) 2015–2030 set the strategic direction for adaptation, emphasizing mainstreaming of climate resilience in all sectors and levels. The National Climate Change Action Plan (NCCAP) 2023–2027 operationalizes these priorities into programs and financing mechanisms, such as the Financing Locally Led Climate Action (FLLoCA) program which channels performance-based climate finance to counties and communities. A National Climate Change Council and Climate Change Directorate provide high-level oversight and technical guidance, while specialized national agencies support implementation: for example, the Kenya Meteorological Department (KMD) provides localized forecasts and early warnings; the National Drought Management Authority (NDMA) leads drought early warning and response; the Kenya Forest Service (KFS) and Kenya Wildlife Service (KWS) support forest conservation, ecosystem restoration, and human–wildlife conflict management; and the National Environment Management Authority (NEMA) ensures environmental safeguards in climate projects. These bodies work in concert with county governments to ensure that national adaptation objectives (e.g. integrated water resource management, climate-smart agriculture, disaster risk reduction, protection of forests and coastal zones) are translated into locally relevant actions.

At the county level, the Kwale County Government serves as the critical bridge between national policies and community needs. The County Government has institutionalized climate change within its structures in line with the Climate Change Act. Notably, the Kwale County Climate Change Act (2022) established a dedicated County Climate Change Unit and a County Climate Change Fund (CCCF) to coordinate climate programs and finance adaptation projects. Climate change is now mainstreamed in the County Integrated Development Plan (CIDP) 2023–2027 and sectoral plans, meaning departments such as Water, Agriculture, Livestock and Fisheries, Roads and Public Works, Health, and Environment must integrate resilience and low-carbon measures into their annual budgets and activities. A County Climate Change Action Plan (2023–2027) further details county-specific targets (e.g. on water security, resilient agriculture, ecosystem restoration, infrastructure protection) and provides an enabling framework for ward-level proposals to access climate finance. County coordination platforms, including the County Steering Group (for development and humanitarian partners coordination) and the County Disaster Management Committee, are increasingly utilized to plan risk reduction and anticipatory actions based on seasonal climate forecasts. Through these mechanisms, the county government aligns external donor/NGO interventions with public investments, avoids duplication, and ensures climate actions are coherent with both national policy and community priorities.

At the ward and community level, implementation and local planning are anchored by Ward Climate Change Planning Committees (WCCPCs) and related community institutions. Each ward (such as Vanga, Ramisi, Pongwe) has established a WCCPC as an inclusive platform for identifying needs, prioritizing adaptation projects, and monitoring progress. The WCCPC is chaired by the Ward Administrator and comprises representatives of local government (e.g. ward Agricultural officers, Public Health officers), community groups (Beach Management Units for fisheries, Water Resource User Associations, Community-Based Organizations), women and youth representatives, persons with disabilities, and respected community leaders. This broad composition mirrors the NAP's emphasis on locally led adaptation and ensures that planning is participatory and considers gender and social inclusion. The WCCPC works closely with issue-specific committees (water user committees manage boreholes and water points; disaster volunteer teams coordinate local emergency response; forest/pasture management groups

oversee ecosystem activities). Historically, many of these community efforts operated in silos or informally; the WCCPC now provides a structured mechanism for coordination at the ward level, facilitating information sharing and joint action across villages and sectors.

To link ward efforts with county support, clear institutional interfaces have been established. A County Climate Change Officer is designated to liaise with each WCCPC, attending ward meetings and ensuring ward priorities are conveyed to the County Climate Change Unit for integration into county programs and CCCF funding pipelines. The Ward Administrator and WCCPC produce quarterly progress reports on adaptation projects, which feed upward to the County Climate Change Unit, capturing outputs, challenges, gender-disaggregated impacts, and resource needs. This reporting routine embeds accountability and allows the county to monitor ward-level progress in line with county and national indicators (e.g. those of FLLoCA and the Climate Change Act). Multi-stakeholder forums are also convened: each ward will hold a biannual Ward Climate Stakeholder Forum under the WCCPC, bringing together county departmental officers, national agencies operating locally, NGOs, CBOs, private sector, and research institutions active in that ward. This forum maps who is doing what and where, aligns technical standards, and resolves any implementation bottlenecks collaboratively. These arrangements ensure vertical integration (ward plans informing county budgets and vice versa) and horizontal coordination (all local actors working in synergy).

2.1 Policy alignment and implementation roles

The CCAAP is anchored in existing policy frameworks. It operationalizes Kenya's NAP 2015–2030 priorities by focusing on the very sectors highlighted at national level, water, agriculture, health, infrastructure, ecosystems, and disaster risk management, but tailors actions to Kwale's local context. It also aligns with the NCCAP 2023–2027, tapping into national programs like FLLoCA and the National Drought Emergency Fund to support local actions. The plan is consistent with the Kwale County Climate Change Action Plan (2023–2027) and the Climate Change Act (2022), leveraging the County Climate Change Fund mechanism to fund community-prioritized projects. Climate change is integrated in the CIDP 2023–2027, meaning this adaptation plan will be mainstreamed into the county's development agenda and annual development plans. Finally, implementation will be guided by institutional roles as defined in law: the WCCPC leading community mobilization and prioritization, county departments providing technical support and funding through sector budgets, and national agencies and development partners providing specialized inputs (early warning information, climate finance, etc.). Table 1 below summarizes the key institutions and their roles in implementing the adaptation actions. The CCAAP explicitly mainstreams Gender Equality and Inclusion (GEI) across all stages of planning and implementation and establishes strong coordination linkages with health, gender, and protection actors to guarantee continuity of essential Sexual and Reproductive Health and Rights (SRHR) and GBV prevention and response services during climate shocks.

Table 1: Institutional Roles in Implementation of CCAAP

Institution / Actor	Roles and Responsibilities
Ward Climate Change Planning Committee (WCCPC)	Lead community engagement, planning and prioritization of ward-level adaptation actions; ensure projects are inclusive (gender and youth); coordinate implementation at ward level.
Kwale County Government Departments	Integrate ward adaptation priorities into sector plans and budgets (Water, Agriculture, Livestock, Fisheries, Roads, Health, Environment); provide technical expertise and co-funding; oversee infrastructure works and service delivery.
County Climate Change Unit (CCU)	Coordinate CCAAP implementation county-wide; manage the County Climate Change Fund (CCCF) allocations;

Institution / Actor	Roles and Responsibilities
	consolidate M&E reporting from wards; liaise with National Climate Change Directorate on finance and reporting compliance.
National Drought Management Authority (NDMA)	Monitor drought conditions; issue early warnings; trigger drought emergency funds for interventions (water trucking, food relief) when thresholds are reached; support community drought contingency planning.
Kenya Meteorological Department (KMD)	Provide localized weather forecasts and seasonal climate outlooks; issue early warnings for extreme weather (heavy rain, storms); support establishment of ward-level weather monitoring (rain gauges) and advisories for farmers and fishers.
Kenya Forest Service (KFS)	Advise on forest restoration and agroforestry; support tree nursery establishment and management (including mangroves); enforce protection of gazetted forests; assist communities in implementing reforestation and woodlot projects.
Kenya Wildlife Service (KWS)	Handle human–wildlife conflict mitigation (e.g. respond to wildlife incursions, problem animal control); support community education on wildlife safety; facilitate compensation claims for wildlife damage; help establish wildlife deterrent measures near parks/reserves.
Beach Management Units (BMUs) & Fisheries Service	Co-manage coastal and marine resources; implement sustainable fishing practices; participate in mangrove conservation and reef protection to buffer against sea-level rise and storms; support alternative livelihoods for fisherfolk (e.g. mariculture).
Community-Based Organizations (CBOs) & Self-Help Groups	Implement community-led projects (water harvesting, tree planting, soil conservation, savings groups); mobilize local resources and labor; ensure inclusion of women, youth, and marginalized in project benefits.
Development Partners & NGOs (e.g. World Bank FLLoCA, UNDP, WWF, Plan International)	Provide funding grants and technical assistance for local adaptation projects; build capacity (training, tools) for communities and county staff; support innovation (e.g. climate-smart agriculture pilots, ecosystem-based adaptation) and document lessons for scaling up.
Private Sector & Cooperatives	Invest in climate-smart value chains (e.g. drought-tolerant seeds, solar irrigation kits); offer products like weather-index insurance for farmers; participate in Public-Private Partnerships for resilient infrastructure (water supply, renewable energy); support community projects through corporate social responsibility.
Academic & Research Institutions (e.g. KMFRI, Universities)	Generate localized climate data and risk analyses; monitor ecosystem changes (marine and terrestrial); pilot new adaptation technologies (e.g. salt-tolerant crops, erosion control methods); evaluate adaptation outcomes and advise on improvements; support climate knowledge management and learning.

3 Chapter 3: Situation analysis of climate vulnerability

3.1 Introduction

Kwale County experiences multiple and interlinked climatic hazards that have intensified over the past decades, directly impacting livelihoods, ecosystems, and infrastructure. The Participatory Climate Vulnerability and Risk Assessments (PCVAs) conducted across fourteen villages namely Anzuwani, Kiwambale, Barabarani, Mkwiro, and Kibuyuni in Pongwe Kikoneni Ward; Mafisini, Dzibwage, Darigube, Nyumba Sita, and Mwagundu in Ramisi Ward; and Mwalewa C, Sega D, Ngweneni, and Mikoroshoni in Vanga Ward, revealed that the county faces recurrent droughts, floods, pests and diseases, sea level rise, environmental degradation, strong winds, and human–wildlife conflicts (HWCs). These hazards are influenced by shifting climate patterns characterized by rising mean annual temperatures (increasing by approximately 0.3°C per decade), irregular and declining rainfall reliability, and prolonged dry spells particularly between July and October. Flood events have become more frequent during the long rains (March–May), leading to displacement and loss of crops and livestock, while extended droughts have intensified water scarcity and food insecurity. Along the coastal belt, sea level rise and coastal erosion threaten settlements and productive areas, with low-lying regions such as Vanga, Kibuyuni, and Mkwiro increasingly experiencing saltwater intrusion and shoreline retreat.

The cumulative effect of these hazards has eroded livelihood security and degraded key ecosystems such as forests, mangroves and kayas thereby increasing competition over scarce resources. The PCVA findings underscore a county-wide trend of increased climate variability where increased frequency of hazards and intensity continue to amplify vulnerabilities across social groups, particularly women, youth and Persons Abled Differently (PAD)necessitating integrated, gender responsive and ecosystem-based adaptation interventions.

3.2 Current climate trends

Kwale County is situated along Kenya's southern coastline within the tropical climatic belt and experiences a bimodal rainfall regime influenced by both the Inter-Tropical Convergence Zone (ITCZ) and the Indian Ocean monsoon systems. The county's climate is predominantly humid along the coastal strip and semi-arid inland, resulting in sharp spatial and temporal variability in rainfall, temperature, and evapotranspiration. Analysis of long-term climate data derived from daily rainfall records (CHIRPS) and evapotranspiration and temperature datasets (ERA5-Ag) reveals a discernible trend of increasing climatic variability, with a tendency toward warmer and more moisture-demanding atmospheric conditions.

Rainfall trends based on CHIRPS datasets indicate an annual mean of approximately 1,200–1,300 mm, with marked spatial heterogeneity across the county's physiographic zones. The PCVAs conducted in fourteen villages across Ramisi, Pongwe Kikoneni, and Vanga confirm these findings, highlighting recurrent drought episodes linked to prolonged dry spells between June and October, as well as localized flood events following intense rainfall, particularly along the Ramisi River basin and coastal depressions near Vanga. This temporal inconsistency has disrupted rain-fed agriculture, reduced groundwater recharge, and increased reliance on boreholes and surface dams. Evapotranspiration dynamics derived from ERA5-Ag datasets show a mean potential evapotranspiration (PET) of approximately 1,600 mm per annum, while actual evapotranspiration (AET) ranges from 1,700 mm to 1,800 mm, depending on land use and vegetation cover (Figure 4). The close alignment of PET and AET values suggests a high atmospheric evaporative demand, leaving little surplus moisture for surface or subsurface storage. Peak PET occurs during July–October when relative humidity drops below 60% and mean daily temperatures exceed 30°C. This persistent atmospheric demand has resulted in soil desiccation, declining vegetation cover, and elevated evapotranspiration rates in degraded landscapes, particularly around the Shimba Hills hinterlands and the semi-arid interior.

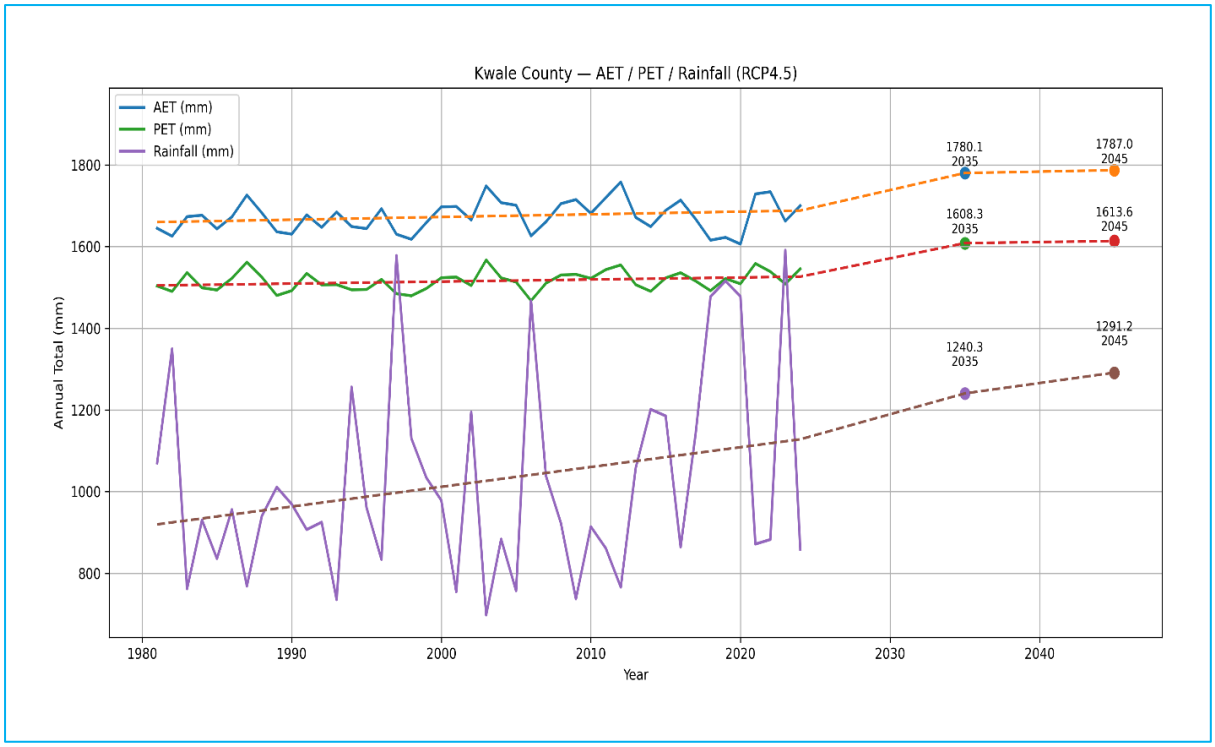

Figure 4: SSP2 4.5 Rainfall, AET and PET Projections (Source: Chirps database and ERA5-Ag, 2022)

Temperature analysis from ERA5-Ag data indicates a mean annual air temperature of approximately 25.5°C and land surface temperatures (LST) between 38°C and 39°C, reflecting significant land–atmosphere heat fluxes (Figure 5). The data show a clear warming trend of about 0.2–0.3°C per decade, with inland areas exhibiting greater thermal anomalies due to reduced vegetation and increased albedo contrast from bare soils. Rising temperatures have heightened evapotranspiration losses, induced heat stress in livestock and crops, and intensified pest and disease outbreaks.

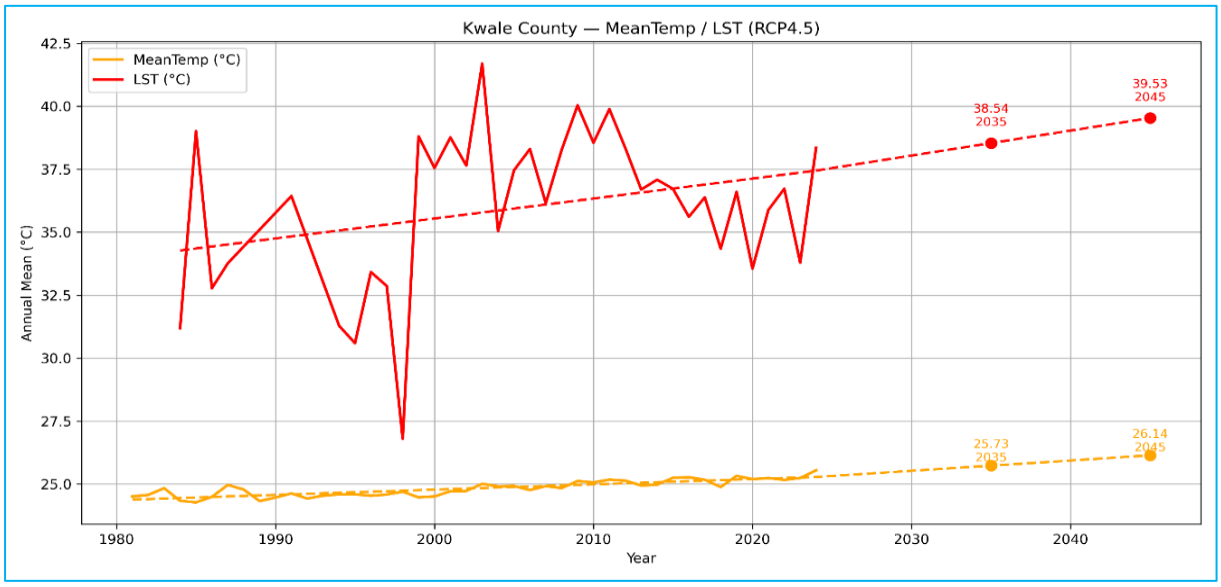

Figure 5: SSP2 4.5 Mean Temperature and Land Surface Temperature Projections (Source: ERA5-Ag database and Landsat 5, 7, 8 & 9 SR ,2022)

3.3 Future climatic trends

Projected climate scenarios for Kwale County, developed from CHIRPS rainfall and ERA5-Ag evapotranspiration and temperature models, suggest an amplification of current trends under both moderate (SSP2–4.5) and high-emission (SSP5–8.5) pathways. Under the SSP2–4.5 moderate-emission scenario, annual rainfall is expected to increase modestly from 1,240.3 mm in 2035 to 1,291.2 mm by 2045, while PET will rise slightly from 1,608.3 mm to 1,613.6 mm, and AET from 1,780.1 mm to 1,787.0 mm (Figure 6). This near-proportional increase across rainfall and evapotranspiration variables indicates a relatively stable hydrological regime with marginal improvement in soil moisture availability. However, the persistence of elevated PET implies sustained atmospheric water demand, limiting the net gains from additional precipitation. Consequently, water stress conditions are projected to persist, particularly in agro-ecological zones dependent on rain-fed production and shallow aquifers.

Under the SSP5–8.5 high-emission scenario, where rapid industrialization, urban expansion, and land-use change drive higher greenhouse gas concentrations, rainfall is projected to rise from 1,299.4 mm in 2035 to 1,352.7 mm in 2045, accompanied by increases in PET (1,684.9 mm to 1,690.4 mm) and AET (1,864.9 mm to 1,872.1 mm). Despite these increases, the balance between rainfall and evapotranspiration remains tight, indicating limited improvement in effective moisture availability. Higher PET and AET under this scenario suggest intensified evaporation and plant transpiration, leading to elevated aridity during the dry season and periodic waterlogging in flood-prone areas during the rains.

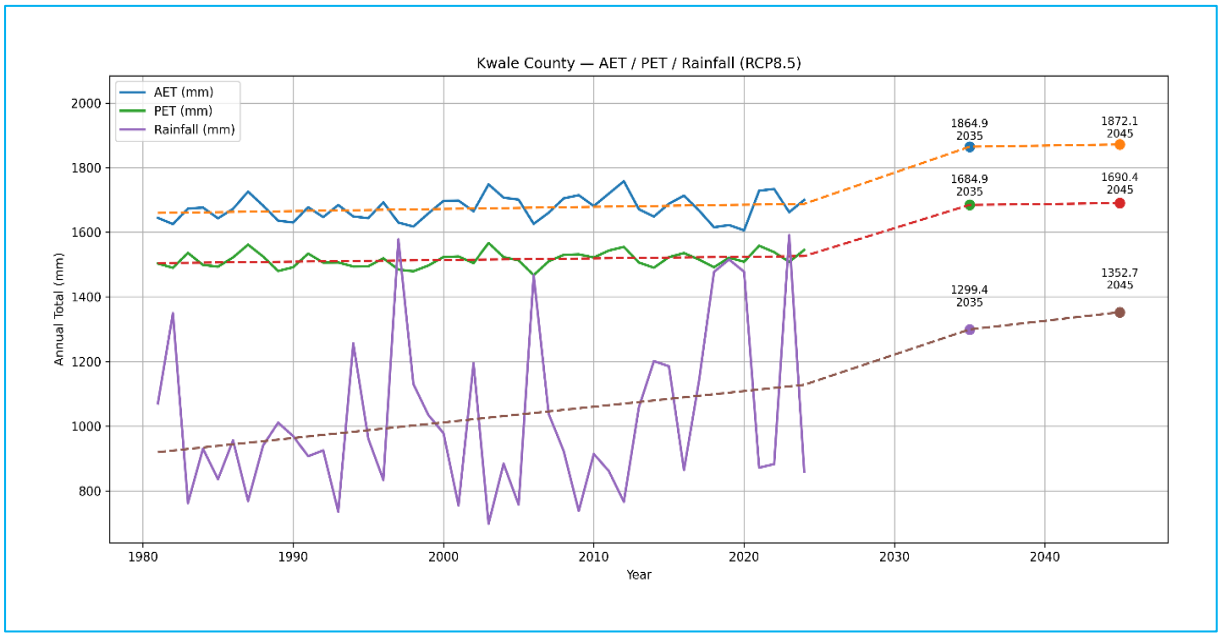

Figure 6: SSP5 8.5 Rainfall, AET and PET Projections. (Source: Chirps database and ERA5-Ag, 2022)

Temperature projections from ERA5-Ag show a consistent warming trajectory. The mean air temperature is projected to increase from 25.73°C to 26.14°C under SSP2–4.5, and from 25.91°C to 26.46°C under SSP5–8.5, while LST will rise from 38.54°C to 39.53°C and 38.70°C to 39.85°C, respectively (Figure 7). The projected warming will heighten the frequency of heatwaves, increase evapotranspiration rates, and exacerbate soil moisture deficits, especially in degraded inland ecosystems. These shifts are likely to alter local hydrological cycles, with implications for agricultural productivity, biodiversity distribution, and human health. Overall, future climatic projections point to a warmer, wetter but more water-stressed environment, underscoring the need for adaptive land-use planning, enhanced water storage infrastructure,

and adoption of climate-smart agricultural and ecosystem-based adaptation interventions to safeguard livelihoods and ecosystems in Kwale County.

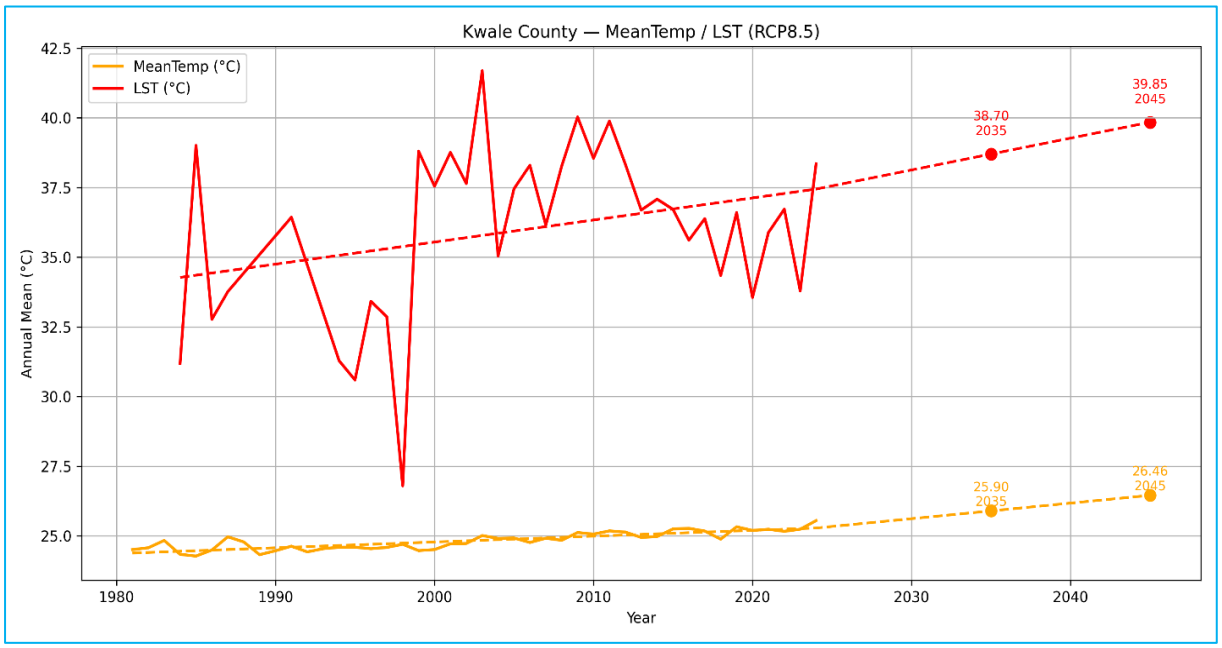

Figure 7: SSP5 8.5 Mean Temperature and Land Surface Temperature Projections. (Source: ERA5-Ag database and Landsat 5, 7, 8 & 9 SR ,2022)

3.4 Climate hazards and vulnerability

3.4.1 Drought

Drought remains the most recurrent and economically disruptive climatic hazard across Kwale County, occurring on average every three to four years. Historical records indicate severe episodes in 1971, 1982, 1992, 2005–2007, 2017, and 2024, each leaving widespread impacts on ecosystems and livelihoods. The increasing frequency and duration of droughts are attributed to erratic rainfall patterns, delayed onset of the long rains, and high evapotranspiration rates driven by rising temperatures. Across the three sampled wards; Pongwe–Kikoneni, Ramisi, and Vanga, prolonged dry spells have depleted freshwater resources, reduced vegetation cover, and intensified land degradation.

Ecologically, recurrent droughts have significantly altered the structure, composition, and functioning of key terrestrial and coastal ecosystems across the county. Forest ecosystems such as Ganda, Kiriku, and Buda have exhibited clear signs of canopy dieback, undergrowth loss, and reduced natural regeneration, thereby undermining their ecological roles in groundwater recharge, soil moisture retention, and microclimate regulation. NDVI (Normalized Difference Vegetation Index) analysis of Buda Forest within Ramisi Ward indicates relatively stable vegetation health, with only minor seasonal variation between dry and wet periods. During the dry season, NDVI values range from 0.26 to 0.52, reflecting moderate vegetation vigor and slight moisture stress resulting from reduced rainfall and elevated evapotranspiration. In the wet season, NDVI values range from 0.24 to 0.53, showing slight improvement in canopy greenness and productivity as soil moisture improves. The minimal seasonal difference underscores the relative resilience of Buda Forest, likely attributable to its closed canopy structure and moisture-retaining soils. However, persistent climatic variability, particularly extended dry spells or delayed rainfall onset, could progressively weaken this ecological stability and compromise long-term forest health.

In contrast, Gonja Forest in Vanga Ward demonstrates greater sensitivity to seasonal moisture fluctuations. NDVI analysis reveals dry-season values ranging between 0.15 and 0.48, indicative of moderate stress due to limited precipitation, high temperatures, and reduced soil moisture. During the wet season, NDVI values rise to 0.24–0.50, suggesting enhanced canopy density, improved regeneration, and increased vegetation productivity following replenished rainfall and nutrient inflow. This pronounced seasonal variation highlights the forest's dependence on consistent precipitation for sustained ecological functioning.

Along the coastal belt, mangrove ecosystems in Vanga, Kiwegu, and the Funzi–Nyumba Sita complex exhibit pronounced vegetation stress during drought periods. NDVI values along the Ramisi–Pongwe Kikoneni–Vanga coastline range from 0.12 to 0.48 in the dry season, reflecting moderate canopy stress caused by high salinity, limited freshwater inflow, and elevated evapotranspiration. During the wet season, NDVI values increase to between 0.14 and 0.55, indicating partial recovery and improved vegetation vigor due to increased rainfall and nutrient availability. However, localized low NDVI values observed between Majoreni and Vanga even during the wet season point to stress factors such as sedimentation and tidal flooding, which inhibit regeneration by smothering pneumatophores and restricting oxygen exchange. These seasonal dynamics underscore the ecological fragility of mangrove systems under shifting climatic conditions, where increasing drought frequency and intensity threaten to exacerbate dieback, seedling mortality, and long-term ecosystem degradation. Similarly, declining baseflows in rivers such as Ramisi, Mwachema, Mwena, and Umba have intensified freshwater scarcity and salinity intrusion in lower reaches, while elevated sea surface temperatures and salinity fluctuations have reduced seaweed productivity in Kibuyuni and Mkwiro. Collectively, these cascading ecological impacts diminish ecosystem resilience, water availability, and the sustainability of coastal livelihoods dependent on natural resource productivity (Source: PCVA, 2025).

3.4.1.1 Exposure to Drought

Drought exposure in Kwale County is widespread and manifests across terrestrial, riparian, and coastal ecosystems. In Pongwe–Kikoneni, rain-fed agricultural zones such as Mabafueni and Mwandeo face frequent crop failure, while settlements around Kiriku Forest suffer from diminishing water supply as shallow wells dry up. In Ramisi, areas such as Mavambe, Fingirika, and Eshu record prolonged soil moisture deficits, reducing maize and cassava yields. In Vanga, communities in Mwalewa C, Ngweneni, and Mikoroshoni experience seasonal water shortages as boreholes run dry, forcing women and children to travel over six kilometres daily in search of water. Forests such as Ganda and Kaya Sega show visible vegetation stress, while the drying of streams feeding the Umba and Ramisi rivers exacerbates saline intrusion into coastal aquifers. The combined reduction of water, vegetation cover, and soil fertility across wards illustrates high countywide exposure. (Source: PCVA, 2025)

3.4.1.2 Socio-Demographic Vulnerability to Drought

Socio-demographic vulnerability to drought is pronounced among women, youth, elderly persons, and PAD whose livelihoods depend on climate-sensitive resources. Women bear the brunt of domestic water scarcity, fuelwood collection, and food production. As household food insecurity rises, they face increased unpaid labour and reduced participation in income-generating activities. Youth engaged in casual farm work and fishing experience loss of income as agricultural productivity declines. Elderly persons and PAD face mobility challenges that limit access to distant water sources and relief services. Economic vulnerability compounds the challenge, most households earn less than KES 10,000 monthly, limiting investment in irrigation or water storage. Weak access to extension services and early warning information further limits preparedness. Across the wards, women's limited land ownership and access to credit hinder

adoption of drought-tolerant crops or livestock breeds, perpetuating gendered vulnerability. (Source: PCVA, 2025)

3.4.2 Floods

Flooding remains one of the most recurrent and destructive hazards in Kwale County, with major flood events recorded in 1961, 1970, 1972, 1997, 2018, and 2023, approximately once every five years. The intensity of floods has increased in recent decades due to erratic rainfall, deforestation, sedimentation, and unregulated settlement in flood-prone areas. Floods affect both inland and coastal systems, with the most severe impacts felt along the Ramisi, Mwachema, Mwena, Mwabanda, Rovu, and Umba rivers, and in low-lying coastal villages such as Bolo, Vingujini, and Nyumba Sita.

Flood events disrupt ecological stability and livelihoods. Forests such as Buda and Kaya Mwabanda experience topsoil erosion and reduced infiltration, diminishing their capacity to regulate hydrological flows. Mangrove ecosystems in Funzi, Vanga, and Shirazi undergo prolonged inundation and sediment accumulation that suffocates pneumatophores and suppresses regeneration. Along the coast, seaweed farms in Kibuyuni, Mwandeo, and Mangawani are dislodged by turbulent floodwaters and sedimentation, reducing yields and labour opportunities for women and youth. Floods also damage rural infrastructure, contaminate water supplies, and spread waterborne diseases. (Source: PCVA, 2025)

3.4.2.1 Exposure to Floods

Exposure is concentrated in the low-lying floodplains, riverbanks, and estuarine zones. In Ramisi, settlements such as Mafisini, Dzibwage, and Darigube face recurrent inundation and saltwater intrusion from the Ramisi River and adjacent channels. In Pongwe–Kikoneni, Mabafueni and Mwandeo experience periodic submersion of croplands due to overflowing tributaries of Mwachema and Mwena. In Vanga, the Umba River floods frequently affect Kiwegu, Jasini, and Vanga villages, destroying farmland and infrastructure. Inadequate drainage and unplanned settlement exacerbate the situation, while the degradation of mangrove buffers and vegetation cover amplifies flood exposure along the coastline. (Source: PCVA, 2025)

3.4.2.2 Socio-Demographic Vulnerability to Floods

Flood vulnerability in Kwale is shaped by socio-economic status, gender, and settlement patterns. Women, particularly those engaged in informal trade and caregiving, experience significant losses during floods due to destruction of property and disruption of household water and sanitation facilities. Children face health risks from contaminated water and unsafe play environments, while elderly persons and PAD face challenges evacuating flooded areas. Poor households, especially those living near riverbanks in makeshift structures, lack financial capacity to rebuild, trapping them in poverty cycles. Female-headed households bear additional responsibilities for food, health, and childcare during recovery. Youth working in fishing and seaweed farming suffer income interruptions as production halts. (Source: PCVA, 2025)

3.4.3 Pests and Diseases

Pest and disease outbreaks are increasingly frequent, occurring roughly every two years and affecting agriculture, livestock, forests, mangroves, and marine ecosystems. The shifts in temperature, humidity, and rainfall patterns have created favourable conditions for pests and pathogens. Historical records link outbreaks in 1992, 2010, 2016, and 2023 to alternating wet–dry cycles across the county.

Agricultural systems face periodic infestations of armyworms, locusts, and cassava mosaic virus, leading to food insecurity. Livestock herds experience tick-borne diseases and foot-and-mouth disease. Forests such as Ganda and Kiriku have suffered pest-induced defoliation, weakening

regeneration. In the marine sector, seaweed farms in Kibuyuni and Mkwiro report frequent outbreaks of "ice-ice" disease and epiphytic infestations, reducing production and affecting women's groups financially. (Source: PCVA, 2025)

3.4.3.1 Exposure to Pests and Diseases

Exposure is high in both inland and coastal systems. Farming areas in Mabafueni, Kiwambale, and Mwalewa C are vulnerable due to monocropping and limited pest surveillance. In Vanga and Ramisi, poor drainage and humid conditions promote pest breeding. Mangrove and seaweed systems are exposed to pathogenic growth following salinity fluctuations. Forest degradation and fragmented habitats reduce ecological resilience against pest invasion, creating a feedback loop that worsens exposure. (Source: PCVA, 2025)

3.4.3.2 Socio-Demographic Vulnerability to Pests and Diseases

Women farmers bear the highest burden due to their role in food production and caregiving. Increased labour demand for pest control and reduced yields elevate time poverty. Poor households lack access to pesticides, veterinary care, and extension services. Elderly persons, children, and PAD are more susceptible to vector-borne diseases such as malaria and dengue during humid seasons. Recurrent outbreaks reduce income, weaken nutrition, and increase household health expenditures. (Source: PCVA, 2025)

3.4.4 Human–Wildlife Conflicts

Human–wildlife conflicts (HWCs) have become a significant and escalating challenge across Kwale County, particularly within the forest–agriculture–settlement interface and along riparian zones. Although these conflicts historically occurred approximately once every eight years, community records and participatory assessments indicate an increase in both frequency and severity since 2023. The expansion of cultivation into wildlife corridors, coupled with habitat fragmentation from logging and charcoal burning, has increased encounters between humans and wildlife in Pongwe–Kikoneni, Ramisi, and Vanga Wards. The main species involved include baboons, monkeys, and vervets raiding crops near forest edges, while crocodile attacks are common along the Ramisi and Umba rivers. Prolonged droughts and water scarcity have further forced wildlife closer to human settlements, escalating competition for natural resources.

3.4.4.1 Exposure to Human–Wildlife Conflicts

Exposure is concentrated in communities situated adjacent to forests, rivers, and water points. In Pongwe–Kikoneni, the most affected villages include Kiwambale, Ganda, and Mzizima, where primate incursions into maize, cassava, and fruit farms have become routine. In Ramisi, households near Buda Forest and Kaya Mwabanda face nocturnal raids from baboons and occasional elephant migration from adjacent wildlife dispersal areas. Along the Umba River in Vanga, crocodile attacks have been documented during water collection and washing activities, especially in the dry seasons when both humans and wildlife depend on shrinking water sources. Habitat degradation, encroachment, and reduced forest buffers amplify these interactions. The destruction of riparian vegetation and overexploitation of forest resources have also diminished natural boundaries that previously separated human and wildlife territories. (Source: PCVA, 2025)

3.4.4.2 Socio-Demographic Vulnerability to HWCs

The socio-demographic vulnerability to HWCs is greatest among poor rural households that depend directly on farming, livestock, and open-access water sources. Women and children are particularly at risk due to their roles in fetching water and tending to farms near forest margins. Injuries and fatalities, although sporadic, have long-term psychosocial and economic consequences. Elderly persons and PAD face the greatest difficulty fleeing or avoiding wildlife encounters due to mobility limitations. Livestock predation and crop destruction significantly

erode household income and food stocks, leading to recurrent food shortages and deepening poverty. The absence of compensation mechanisms under national wildlife policy frameworks further discourages reporting and fosters retaliatory killings, which degrade ecological balance and reduce biodiversity in already fragile habitats. (Source: PCVA, 2025)

3.4.5 Strong Winds

Strong winds are an increasingly destructive hazard across Kwale's coastal belt, occurring primarily during the southeast (Kusi) and northeast (Kaskazi) monsoon transitions, roughly every four years. Historical observations highlight damaging events in 1984, 1998, 2014, 2019, and 2023. Wind speeds along the coast can reach between 25 and 35 knots (46–65 km/h), causing extensive damage to coastal infrastructure and vegetation. These winds are often accompanied by storm surges and heavy rainfall, intensifying impacts on fishing communities, marine ecosystems, and built infrastructure.

3.4.5.1 Exposure to Strong Winds

Exposure is pronounced in coastal and near-coastal zones where settlement expansion has occurred without adequate windbreaks or vegetative buffers. In Pongwe–Kikoneni, villages such as Mkwiro and Kibuyuni are severely affected, with strong gusts uprooting mangroves, destroying seaweed racks, and damaging fish landing sites. In Vanga, Sega D and Kiwegu face recurrent destruction of roofs and community structures, while Ramisi's coastal settlements, particularly Mwangundu, experience repeated loss of makuti-roofed housing and schools. Inland, deforestation in Kiriku and Ganda Forests has diminished the natural wind shielding effect, exposing agricultural fields and homesteads to greater turbulence and dust storms. The loss of vegetative cover and coastal vegetation has also intensified salt spray, which corrodes metal roofing and infrastructure. (Source: PCVA, 2025)

3.4.5.2 Socio-Demographic Vulnerability to Strong Winds

Women and youth whose livelihoods depend on marine and coastal activities are highly vulnerable to wind hazards. Seaweed farming groups, primarily composed of women, experience total production loss when cultivation structures are destroyed. Fishers operating small, unpowered boats along the nearshore face increased risks during rough sea conditions, often leading to temporary livelihood suspension. Female-headed households, whose homes are predominantly made from lightweight materials, incur recurrent losses from structural damage. PAD and elderly residents face heightened danger during evacuations, while children's education is disrupted when schools are damaged or converted into temporary shelters. The economic costs of reconstruction and the absence of insurance mechanisms delay recovery, particularly for low-income households. Women also report heightened exposure to gender-based violence in overcrowded temporary shelters established after windstorms. (Source: PCVA, 2025)

3.4.6 Environmental Degradation

Environmental degradation is both a climate hazard and a key driver of vulnerability in Kwale County, amplifying the impacts of droughts, floods, and coastal erosion. The degradation is largely anthropogenic, stemming from unsustainable land use, deforestation, mangrove cutting, charcoal burning, and sand mining. Over the last three decades, satellite assessments and field surveys have documented significant reduction in forest cover, sedimentation of rivers, and loss of mangrove ecosystems along the South Coast. This has weakened the county's ecological resilience, reducing its natural capacity to buffer climatic shocks.

3.4.6.1 Exposure to Environmental Degradation

Exposure is concentrated in critical ecosystems and community landscapes where degradation processes are most active. In Vanga Ward, the Gonja and Kaya Sega forests are heavily degraded

due to overharvesting of timber and charcoal production, leading to soil erosion, loss of biodiversity, and diminished water infiltration. In Ramisi, sand mining along the Ramisi, Mwabanda, and Rukumbwi rivers has destabilized riverbanks and increased sediment loads downstream, contributing to siltation of mangrove areas in Funzi and Shirazi. In Pongwe–Kikoneni, deforestation of Ganda and Kiriku Forests for agricultural expansion has altered hydrological balance, increasing surface runoff and reducing groundwater recharge. The degradation of mangroves in Vanga and Kiwegu has further exposed coastal communities to wave action, saltwater intrusion, and erosion. (Source: PCVA, 2025)

3.4.6.2 Socio-Demographic Vulnerability to Environmental Degradation

Communities living adjacent to degraded ecosystems bear the greatest social and economic burden. Women and youth engaged in fuelwood collection, charcoal production, and mangrove harvesting experience declining incomes as resources become scarce. The degradation of riverine vegetation increases the distance and time women must travel to collect water and firewood, intensifying time poverty and exposure to physical risks. Youth engaged in sand mining and timber extraction are trapped in unsafe, low-paying jobs with limited future prospects. The elderly and PAD, who depend on proximate natural resources for food, medicine, and materials, face increased hardship as resource availability diminishes. The cumulative effect of environmental degradation is a cycle of declining productivity, ecosystem fragility, and livelihood vulnerability that undermines long-term resilience across the county. (Source: PCVA, 2025)

3.4.7 Sea Level Rise

Sea level rise is an emerging but rapidly intensifying hazard along Kwale's coastline. Observations from the Kenya Meteorological Department and community monitoring indicate a steady increase of approximately 3.0–3.5 mm per year since 2015, consistent with global oceanographic trends. The phenomenon is compounded by land subsidence, coral reef degradation, and declining mangrove cover. Recurrent tidal flooding, saline intrusion, and coastal erosion have been reported across the county's shoreline, particularly in Mkwiro, Kibuyuni, Vanga, and Kiwegu.

3.4.7.1 Exposure to Sea Level Rise

Exposure is concentrated in low-elevation coastal settlements, estuaries, and mangrove fringes. In Pongwe–Kikoneni, Mkwiro and Kibuyuni villages experience periodic inundation during high tides, with seawater encroaching into homesteads, farmlands, and community graveyards. The loss of mangrove cover has weakened the natural coastal defense system, accelerating shoreline retreat. In Vanga and Kiwegu, rising sea levels have caused saline intrusion into freshwater wells and soils, rendering them unsuitable for agriculture. Coastal aquifers are increasingly brackish, affecting domestic water quality. In Ramisi, Funzi Island and nearby mangrove zones have seen submersion of previously productive areas, threatening fish breeding grounds. The encroachment of the sea has also damaged coastal infrastructure, including fish landing sites and access roads critical for trade. (Source: PCVA, 2025)

3.4.7.2 Socio-Demographic Vulnerability to Sea Level Rise

The socio-economic impacts of sea level rise are profound and multidimensional. Women's seaweed cooperatives in Kibuyuni and Mkwiro suffer recurring losses as high tides destroy cultivation lines and drying racks, directly reducing household income. Artisanal fishers lose access to landing beaches and face declining fish catches as breeding grounds are inundated. Low-income households near the coast face repeated displacement, with temporary relocations causing social disruption and increased vulnerability to disease and violence. Salinity intrusion reduces the productivity of kitchen gardens and freshwater availability, worsening food insecurity and forcing women to spend more time sourcing potable water. Cultural heritage

sites, including graveyards and sacred kaya forests near the shoreline, are at risk of irreversible loss. The absence of relocation plans and limited financial capacity constrain adaptation, leaving coastal communities highly exposed. (Source: PCVA, 2025)

3.5 Classification of Exposure and Resilience through Livelihoods

The County's livelihood systems are deeply intertwined with its natural resource base, making them highly exposed to environmental, economic, and institutional shocks. Despite significant ecological endowments spanning marine, coastal, and inland ecosystems, the capacity for resilience remains constrained by low income diversity, high poverty levels, and limited access to productive assets. According to county data, the overall poverty headcount stands at approximately 47%, well above the national average, reflecting widespread structural vulnerability. The livelihoods that sustain the majority of rural and peri-urban households are predominantly informal and climate-sensitive, including fishing, small-scale agriculture, tourism, seaweed farming, livestock production, beekeeping, and forest-based enterprises. Income stability across these sectors is increasingly affected by droughts, floods, sea-level rise, and ecosystem degradation, thereby necessitating a countywide classification of exposure and resilience through the livelihood lens. (Source: PCVA, 2025)

3.5.1 Livelihood Structure and Monthly Incomes

Kwale County's economy is diverse yet predominantly rural, with livelihoods reflecting ecological zones and proximity to markets. Along the coastline in Vanga, Ramisi, and Pongwe–Kikoneni wards, fishing remains the dominant livelihood, providing employment to thousands of artisanal fishers and fish processors. Monthly incomes range widely, from KES 15,000 for small-scale fishers in shallow waters to KES 70,000 or more for those with access to deeper offshore fishing grounds and established market linkages in Shimoni and Diani. Seaweed farming, primarily undertaken by women's cooperatives in Kibuyuni, Mkwiro, and Vanga, generates modest but steady earnings averaging KES 3,000–6,000 per month and plays a key role in supporting household nutrition, women's empowerment, and coastal ecosystem stewardship.

Agriculture remains the mainstay for inland and peri-coastal communities, particularly in Ramisi. The sector is dominated by rain-fed maize, cassava, sorghum, green grams, and horticultural crops such as tomatoes and watermelons. Monthly income from small-scale farming averages KES 10,000–25,000, though yield variability is high due to erratic rainfall and recurrent pest infestations. Livestock keeping, mainly cattle, goats, and poultry, is practiced in drier inland areas with typical monthly returns ranging between KES 5,000 and KES 20,000, depending on herd size and market access.

Forest-based enterprises such as the sale of mangrove seedlings, beekeeping, and sustainable timber harvesting are growing income sources in coastal wards. Communities in Ramisi, Vanga, and parts of Pongwe Kikoneni earn between KES 20,000 and KES 50,000 per month from mangrove seedling sales for restoration projects, while beekeeping initiatives in both coastal and inland areas such as Pongwe Kikoneni yield up to KES 80,000 per harvest cycle, making apiculture one of the most climate-resilient emerging livelihoods. Youth-driven activities such as motorbike transport (boda-boda) and casual construction work provide crucial income diversification, generating KES 10,000–KES 40,000 per month depending on seasonality and road conditions.

Tourism remains another major income earner, especially in areas adjacent to beach corridors such as Mkwiro. Households engaged in tourism-related activities, curio vending, boat transport, snorkeling, and hospitality, earn between KES 40,000 and KES 80,000 monthly during peak seasons. Small-scale businesses, including kiosks, tailoring, and vending of

agricultural produce, contribute KES 5,000–20,000 monthly and are critical for urban and peri-urban women traders. The presence of informal savings and credit groups (chamas), government cash transfers, and occasional remittances provides a modest but important financial safety net, averaging KES 2,000 per household monthly. Collectively, these income streams underscore the importance of diversified and gender-responsive livelihood systems to strengthen resilience at both household and county levels. (Source: PCVA, 2025)

3.5.2 Vulnerability of livelihood options

Livelihoods across Kwale County exhibit a high degree of vulnerability to climate hazards due to their heavy reliance on natural resources, limited technological innovation and weak adaptive infrastructure. The county's diverse economy, spanning marine, coastal, and terrestrial zones, is underpinned by livelihoods that are intricately linked to the ecological integrity of forests, rivers, mangroves, coral reefs, and arable land. These ecosystems are increasingly destabilized by recurrent droughts, floods, sea-level rise, and extreme weather events, exposing households to compounding climate risks and economic fragility. The capacity of communities to recover from shocks remains low, constrained by persistent poverty, inadequate social protection systems, and limited diversification beyond resource-dependent sectors.

Fishing, which forms the backbone of the coastal economy in Vanga, Ramisi, and Pongwe–Kikoneni, is among the most climate-sensitive livelihoods in the county. The combined effects of rising sea surface temperatures, ocean acidification, and unpredictable rainfall patterns are altering fish migratory routes and breeding cycles, leading to declining catches. Increasing frequency of storm surges, strong winds, and tidal variability shortens fishing days and heightens occupational risks for artisanal fishers, while wave action and coral bleaching degrade spawning grounds and reduce species diversity. The destruction of landing sites, canoes, and gear during monsoon storms further increases production costs. These biophysical pressures are compounded by socio-economic vulnerabilities such as lack of cold storage, weak market linkages, and absence of insurance for fishing assets. For small-scale fishers, particularly women engaged in post-harvest processing and fish vending, climate shocks translate directly into income losses, food insecurity, and debt accumulation.

Seaweed farming, largely practiced by women's groups in Kibuyuni, Mkwiro, and Vanga, exemplifies gendered vulnerability to climate change. The activity is highly dependent on stable sea temperatures, salinity, and water clarity, all of which are being disrupted by fluctuating ocean conditions. Prolonged droughts increase evaporation and raise salinity levels, while heavy rainfall introduces sediment runoff that smothers seaweed and reduces light penetration needed for photosynthesis. During storm seasons, waves destroy racks and lines, forcing temporary livelihood suspension. The limited access to alternative cultivation sites, processing technology, and adaptive research exacerbates production losses. Women's restricted access to credit and equipment further constrains recovery, making seaweed farming one of the most climate-exposed and economically fragile value chains in the blue economy.

Agriculture and livestock production, which sustain the majority of rural households Mafisini and Pongwe Kikoneni, face multi-hazard vulnerability driven by rainfall variability, soil degradation, and pest infestations. Erratic rainfall patterns and recurrent droughts cause widespread crop failures, particularly for maize, cassava, and sorghum, which form the dietary staples. Extended dry periods deplete soil moisture, accelerate nutrient loss, and increase dependence on costly irrigation or relief assistance. In contrast, intense rainfall and floods wash away topsoil, destroy crops, and increase pest populations such as the fall armyworm. Livestock systems are similarly at risk, prolonged droughts deplete pasture and water, increase disease outbreaks, and cause livestock mortality. Women and children often bear the brunt of these stresses, spending long hours herding animals and fetching water during dry seasons. The

predominance of rain-fed farming, lack of climate-smart technologies, and poor access to veterinary services reduce resilience, reinforcing cycles of food insecurity and poverty.

Forest-based livelihoods such as beekeeping, mangrove seedling production, and the sustainable harvesting of forest products offer moderate climate resilience but remain indirectly vulnerable to environmental degradation. The decline of forest cover in Ganda, Kiriku, Buda, and Gonja Forests has reduced habitat diversity and disrupted ecological services essential for pollination and honey production. Beekeeping is directly affected by the reduction of flowering plants during prolonged dry seasons, deforestation, and pesticide use, which lower nectar flow and colony survival. Similarly, mangrove-based livelihoods, though vital for restoration and conservation, are constrained by extreme weather conditions; droughts inhibit propagule germination while floods and sea-level rise wash away nurseries. The loss of mangrove cover also undermines shoreline stability, further exposing these enterprises to tidal damage. Despite their environmental benefits, the financial and institutional support for these community-based enterprises remains limited, restricting their ability to scale up and adapt to changing climatic conditions.

Tourism, a key economic pillar concentrated in Mkwiro and Kibuyuni faces long-term climate risks that threaten both the natural and economic capital of the county. Coral bleaching, coastal erosion, and the degradation of marine biodiversity diminish the aesthetic and recreational appeal of tourist destinations. Rising sea levels and frequent storm surges threaten beach infrastructure, hotels, and marine parks, leading to seasonal business disruptions and reduced visitor arrivals. As a result, households dependent on tourism-related activities, such as curio vending, boat tours, and hospitality, experience income volatility and job insecurity. Climate-related ecosystem degradation also reduces the attractiveness of nature-based tourism, which forms the foundation of Kwale's ecotourism model.

Informal livelihoods such as motorbike transport (boda-boda), casual labour, and small-scale vending, though less directly climate-dependent, are indirectly vulnerable to extreme weather and economic shocks triggered by climate variability. Heavy rainfall and flooding render roads impassable, restrict mobility, and disrupt local trade networks, while prolonged droughts elevate temperatures, reduce customer turnout, and increase fuel costs. Seasonal fluctuations in agricultural productivity and market prices further affect purchasing power, leading to income instability among traders and transport operators. For youth and women engaged in these sectors, the lack of savings, formal insurance, and social protection mechanisms heightens exposure to income shocks, perpetuating livelihood insecurity.

3.5.3 Overall Classification and Implications

The overall classification positions Kwale County as a high-exposure and low-to-moderate-resilience system, where livelihoods remain deeply dependent on climate-sensitive resources and ecosystems. The dual pressures of environmental degradation and socio-economic inequality continue to erode adaptive capacity. Coastal and inland households alike are increasingly exposed to income volatility from droughts, floods, sea-level rise, and pests, while structural barriers i.e., limited access to finance, weak infrastructure, and entrenched gender disparities, further constrain resilience.

To build long-term resilience, Kwale County must prioritize diversification and value addition within key sectors such as fisheries, mangrove-based enterprises, agroforestry, beekeeping, and climate-smart agriculture. Scaling up microfinance and cooperative savings platforms, strengthening youth and women's enterprise programs, and improving access to credit will help households withstand climate shocks. Additionally, investment in ecosystem-based adaptation

(EbA), including mangrove and forest restoration, catchment rehabilitation, and sustainable land use management, will enhance ecological stability.

Complementary improvements in rural infrastructure, such as feeder roads, cold storage, irrigation systems, and renewable energy access, will reduce transaction costs and strengthen value chains. Finally, integrating gender-responsive planning, social protection measures, and inclusive capacity-building within the County Climate Change Adaptation Action Plan will enable Kwale's communities to transition from reactive coping toward proactive, sustainable, and equitable climate resilience. (Source: PCVA, 2025)

3.6 Analyzing causes of vulnerability

3.6.1 Classification of Severity of Vulnerabilities and Their Causes

The county exhibits differentiated levels of vulnerability to climate hazards, driven by ecological location, livelihood dependence, socio-economic status, and institutional capacity. The classification of vulnerability across the county's wards reveals distinct spatial patterns: coastal areas such as Vanga, Ramisi, and Pongwe–Kikoneni exhibit high vulnerability, primarily due to their proximity to the Indian Ocean, dependence on marine and mangrove ecosystems, and exposure to saline intrusion, sea-level rise, and flooding. In contrast, inland areas such as Ramisi display moderate to high vulnerability, driven by recurrent droughts, pest infestations, and limited water accessibility that threaten agricultural and livestock-based livelihoods.

The severity of vulnerability is further compounded by widespread poverty, inadequate livelihood diversification, and weak adaptive infrastructure. Households dependent on climate-sensitive sectors; fishing, rain-fed farming, livestock rearing, and mangrove harvesting, experience recurrent income shocks and food insecurity. Women remain disproportionately vulnerable due to their concentration in low-income and resource-dependent activities such as small-scale vending, seaweed farming, and mangrove seedling production. Their limited access to credit, training, and productive assets constrains recovery during climate crises. Persons abled differently (PAD), elderly persons, and children are also among the most affected groups due to mobility limitations and dependency on others for essential services.

3.6.2 Identification of Unsafe Conditions

Unsafe conditions across Kwale County reflect the immediate environmental and socio-economic circumstances that heighten exposure to climate-related hazards. Along the coastline, settlements in low-lying zones such as Vanga, Mkwiro and Kibuyuni experience recurrent tidal flooding, saline intrusion, and shoreline erosion, leading to frequent displacement, infrastructure loss, and contamination of freshwater wells. Inadequate drainage systems, poor waste management, and encroachment into mangrove belts have exacerbated flood impacts and accelerated coastal degradation. Inland, communities in Ramisi face unsafe conditions arising from water scarcity, land degradation, and deforestation, which increase vulnerability to drought and soil erosion.

Gendered vulnerabilities are prominent within these unsafe environments. Women and girls often traverse degraded landscapes to fetch water and firewood, exposing them to physical and health risks, particularly during dry seasons. The absence of climate-resilient infrastructure, such as elevated housing, secure fish landing sites, water pans, and all-weather roads, limits the safety and mobility of vulnerable populations. Households headed by widows, elderly persons, and PAD face disproportionate exposure due to weak housing structures and lack of resources to relocate from hazard-prone zones. (Source: PCVA, 2025)

3.6.3 Identification of Dynamic Pressures or Determinants of Vulnerability

Dynamic pressures in Kwale County refer to the processes that convert underlying weaknesses into direct unsafe conditions, amplifying exposure to climate hazards. These pressures are both ecological and socio-economic in nature, stemming from rapid population growth, unregulated land use, overexploitation of natural resources, and weak policy enforcement. Across the county, expanding settlements and infrastructure development have encroached on ecologically fragile zones such as mangroves, forests, wetlands, and riparian buffers.

Forests such as Ganda, Buda, and Kiriku, together with mangrove systems in Vanga, Funzi, and Shirazi, show significant degradation from fuelwood harvesting, charcoal production, and unsustainable logging. These activities diminish vegetative cover, disrupt biodiversity, and reduce the natural buffering capacity of ecosystems against floods and strong winds. For instance, recent assessments (PCVA, 2025) indicate that approximately one-third of mangrove areas in Mkwiro and Kibuyuni are degraded due to human pressure and sedimentation from upstream catchments. The ecological consequences include reduced fish nursery habitats, increased shoreline erosion, and weakened carbon sequestration potential.

Dynamic pressures are equally evident in the agricultural zones of Ramisi and Vanga, where extensive grazing, poor soil management, and deforestation have led to declining fertility and reduced water infiltration. These trends have triggered a feedback loop of resource depletion, forcing households to engage in short-term coping mechanisms such as charcoal production, sand harvesting, and informal mining, all of which further degrade ecosystems and increase vulnerability to climate hazards.

Institutionally, fragmented coordination between county departments, local committees, and national agencies undermines integrated resource governance. Enforcement of environmental by-laws is inconsistent, and early warning systems for droughts and floods are either weakly operational or entirely absent in remote wards. The lack of real-time climate information reduces community preparedness, while fluctuating market prices for agricultural and marine products further erode livelihood stability. Socially, the exclusion of women and youth from decision-making bodies, especially within Beach Management Units (BMUs), Community Forest Associations, and County Climate Change Planning Committees, reduces the inclusivity and sustainability of adaptation efforts. (Source: PCVA, 2025)

3.6.4 Identification of Underlying Causes of Vulnerability

The underlying causes of vulnerability in Kwale County are rooted in the intersection of poverty, environmental degradation, weak governance, and gender inequality. Structural poverty remains pervasive across all six sub-counties, with the majority of households depending on subsistence farming, artisanal fishing, and informal trade. Low incomes constrain access to education, technology, and financial services, perpetuating dependence on climate-sensitive livelihoods. According to PCVA (2025), the average household size in Kwale ranges between six and nine members, contributing to high dependency ratios and increasing pressure on land, water, and forest resources.

Environmental degradation is a key underlying driver, manifested through mangrove deforestation, forest loss, soil erosion, and marine ecosystem decline. The degradation of critical ecosystems such as the Ramisi River catchment, the Umba Basin, and the Ganda–Kiriku forest complex has reduced water retention capacity and increased susceptibility to floods, droughts, and storm surges. The loss of vegetative cover not only undermines biodiversity but also weakens the ecological functions essential for soil stabilization, groundwater recharge, and local climate regulation.

Institutional weaknesses exacerbate these vulnerabilities. Limited financial and human resources constrain the County Government of Kwale's capacity to enforce environmental regulations, mainstream climate change into development planning, and coordinate sectoral adaptation efforts. Gaps in inter-departmental coordination between climate change, agriculture, fisheries, water, and environment departments delay policy implementation and reduce efficiency. The absence of localized early warning and response mechanisms results in delayed disaster preparedness and reactive crisis management rather than proactive adaptation.

Gender inequality further compounds vulnerability at both household and community levels. Customary land tenure systems and patriarchal norms restrict women's ownership of productive assets and limit their participation in decision-making structures. Consequently, women are overrepresented in low-return, climate-sensitive livelihoods, particularly in agriculture, seaweed farming, and small-scale vending. This economic marginalization reduces their capacity to invest in adaptive measures or diversify income sources. Gendered barriers also impede access to training, climate information, and credit, perpetuating structural inequities that weaken overall community resilience. (Source: PCVA, 2025)

3.6.5 Prioritization of Causes

The prioritization of vulnerability causes across Kwale County, based on participatory ranking and cross-ward consultations, highlights poverty, environmental degradation, and weak institutional capacity as the top-tier drivers of vulnerability. Poverty limits the ability of households to invest in adaptation technologies, diversify livelihoods, or recover from climate shocks. Environmental degradation diminishes the county's natural defense systems, undermining ecosystem services critical for climate regulation and disaster mitigation. Weak institutional frameworks, characterized by limited funding, fragmented coordination, and insufficient enforcement of climate-related policies, delay the operationalization of adaptation strategies. Medium-priority causes include gender inequality, population growth, and limited access to climate information, all of which influence vulnerability dynamics across different socio-ecological zones. Gender disparities restrict adaptive decision-making and constrain equitable participation in resource governance. Rapid population growth and settlement expansion exacerbate land scarcity and resource competition, while the lack of accessible, timely climate information hinders local-level planning and preparedness.

Addressing these root causes requires an integrated and multi-sectoral approach anchored in ecosystem restoration, livelihood diversification, and institutional strengthening. Scaling up community-based adaptation (CBA) initiatives, such as mangrove and forest restoration, water harvesting, and climate-smart agriculture, can reduce exposure and enhance ecological resilience. Concurrently, promoting gender-responsive financing mechanisms, microcredit for women-led enterprises, and inclusive early warning systems will improve socio-economic resilience. Strengthening coordination among county departments, local committees, and development partners is essential to ensure that climate action in Kwale transitions from fragmented interventions to cohesive, sustainable, and equitable adaptation outcomes. (Source: PCVA, 2025)

4 Chapter 4: Gender/Age Vulnerability Analysis for Climate Hazards

Gender and age profoundly shape the degree of exposure, sensitivity, and adaptive capacity of different population groups in Kwale County. Climate hazards, including drought, floods, pests and diseases, human–wildlife conflict, environmental degradation, sea level rise, strong winds, and coral bleaching, intersect with deeply entrenched gender inequalities, labour divisions, and socio-economic disparities (Table 2). Women and girls bear a disproportionate burden of these risks due to socially ascribed roles, restricted access to productive assets, and their overrepresentation in climate-sensitive livelihood systems such as small-scale farming, artisanal fisheries, and informal natural resource use (UN Women, 2022; IPCC, 2022).

A defining feature of vulnerability among women and girls in Kwale is the heavy load of unpaid domestic and care work. Across the counties, women undertake extensive daily tasks including laundry, cooking, fetching water, collecting firewood, sweeping, preparing and bathing children, assisting with homework, and performing farm work. When economically valued during PCVAs, this unpaid workload ranged from KES 1,500–2,000 per day, contributing significantly to household functioning yet remaining invisible in formal economic statistics. Nationally, unpaid care work represents a major economic contribution, with Kenyan women performing over 70% of such labour (KNBS, 2023; UNDP, 2022). Climate hazards intensify this workload by increasing time spent on resource collection, household management, and caregiving during crises. This "time poverty" constrains women's opportunities for income generation, participation in adaptation planning, and access to education and training (ILO, 2018; UN Women, 2023).

Drought exacerbates gendered labour inequalities as drying water sources force women and girls to walk longer distances, sometimes over 5 kilometres, to fetch water, gather firewood, and secure food. Such conditions increase fatigue, musculoskeletal strain, dehydration, and psychosocial stress (UNICEF, 2021). Female-headed households are particularly vulnerable due to limited labour support and restricted access to financial resources. Men face loss of agricultural productivity, livestock mortality, and declining forest resources, which weaken household income and heighten stress (FAO, 2022). Youth (18–35) experience reduced access to casual labour, farm-based apprenticeships, and value-chain opportunities, increasing idleness and exposure to unsafe coping strategies (UNFPA, 2022). For the elderly and Persons Abled Differently (PADs), mobility challenges restrict access to water, food, and relief supplies, making them heavily dependent on family or caregivers during prolonged droughts (WHO, 2021).

Floods intensify caregiving burdens for women, who often manage sick or displaced family members, salvage household assets, and secure food and water when supply chains are disrupted. Loss of informal trading spaces, such as local markets, directly reduces women's income. Flood-related contamination of wells and boreholes forces women and girls to travel farther or resort to unsafe sources, increasing exposure to disease (UNICEF, 2020). Men disproportionately lose income from fishing, transport, and daily wage labour when infrastructure is damaged. Youth face repeated school closures, diminished income opportunities, and increased vulnerability to exploitation. Elderly people and PADs experience high risk during evacuation due to restricted mobility, dependence on caregivers, and limited access to essential medicines and safe shelter (WHO, 2020).

Pests and diseases affect women through reduced productivity in subsistence agriculture, small livestock rearing, and seaweed farming. Their restricted access to certified seeds, extension services, credit, and pest management technologies lowers their adaptive capacity (FAO, 2021). Men incur income losses from failed crops and livestock diseases, while youth lose farm-related casual labour opportunities. Elderly persons and PADs experience heightened food insecurity and greater dependency when household food stocks decline.

Human–wildlife conflict (HWC), especially involving elephants, baboons, and crocodiles, reduces women's mobility and restricts activities such as firewood collection, water fetching, and small-scale trade, heightening fear and productivity loss (UNEP, 2022). Women also face barriers accessing compensation schemes due to limited documentation and decision-making power. Men engaged in farming and livestock rearing bear financial losses and risk injury while guarding farms. Youth face similar risks, while elderly persons and PADs are extremely vulnerable due to slow mobility and inability to flee during wildlife encounters.

Environmental degradation, including loss of mangroves and forest resources, significantly intensifies women's workload. As firewood, wild foods, and herbs become scarce, women walk longer distances, sometimes exceeding five kilometres daily, to meet basic household needs. This reduces their time for childcare, rest, and income-generating activities and affects their health. Men face reduced fish catch and must travel farther offshore, increasing operational costs and safety risks. Youth lose opportunities in nature-based enterprises such as bee-keeping, seedling production, and eco-tourism. Elderly persons and PADs experience increased dependence and reduced access to degraded landscapes.

Sea level rise disproportionately affects women involved in seaweed farming and shoreline trade. Destruction of drying racks, seaweed lines, and marketing spaces reduces their economic independence (UN Women, 2021). Women bear the burden of relocating seaweed lines, often under dangerous tidal conditions. Men face damaged boats, submerged landing sites, and reduced fish stocks, while youth lose tourism-related opportunities. Elderly persons and PADs face limited relocation ability, heightened exposure to contamination, and emotional distress associated with displacement. Strong winds increase the caregiving burden on women, who must secure household belongings, care for injured family members, and manage disruptions in domestic routines. Men, especially fishermen, are at risk of capsizing, gear loss, and accidents. Youth face physical danger from collapsing structures, while elderly individuals and PADs face significant challenges seeking safe shelter.

Gender-based violence (GBV) further compounds these gendered vulnerabilities, functioning both as a climate-induced risk and as a structural barrier to resilience in Kwale County. Emerging global evidence demonstrates a strong association between climate stressors and rising GBV incidence, with every 1°C increase in temperature linked to a 4.7% rise in intimate partner violence (UN Spotlight Initiative, 2024). Heatwaves have been associated with a 28% increase in femicide, while climate-driven displacement heightens risks of child marriage, sexual exploitation, and trafficking (UN Spotlight Initiative, 2024). In Kenya, rising temperatures and socioeconomic stressors have coincided with escalating femicides with over 930 cases documented between 2016 and 2024, highlighting how climate stress intersects with entrenched gender inequalities (Kenya Femicide Data Study, 2024). These dynamics are most acute in fragile, resource-dependent communities such as Kwale, where drought, floods, and livelihood loss amplify household tensions, reduce economic security, and disproportionately expose women and girls to physical, psychological, and sexual violence. GBV undermines climate resilience by restricting women's mobility, decision-making autonomy, access to productive assets, and participation in county and ward-level adaptation processes (UN Women, 2022). As such, GBV must be integrated into anticipatory climate action, beyond response and preparedness, through early warning systems, forecast-based triggers, community surveillance, and pre-positioned support services coordinated with mandated county structures such as the County Gender Department, Social Services, child protection units, and GBV coordination mechanisms that programmes like COSME have actively engaged. Embedding GBV mitigation within resilience planning ensures that climate adaptation measures safeguard rights, safety, and justice, which are foundational to sustainable and inclusive resilience building.

Table 2: Gender/Age Vulnerability Analysis for Climate Hazards

Women	Men	Youth	Elderly	PADs
Drought				
• Female-headed households bear highest burden of water, food, and fuelwood provision • Walk longer distances to depleted water sources → fatigue, dehydration, back strain • Reduced time for childcare, rest, and income generation • Increased psychological stress due to food shortages	• Livelihood insecurity from reduced crop yields, livestock productivity, and forest resources • Increased financial pressure and stress • Reduced availability of agricultural labour opportunities	• Reduced productivity and income from casual labour losses • High idleness → increased risk of crime, drugs, and unsafe coping strategies • Higher incidence of early marriages and pregnancies among girls	• Mobility limitations restrict access to water and food • Increased susceptibility to heat stress, dehydration, and hunger • Dependence on caregivers worsens during drought periods	• High dependency for water, food, and healthcare access • Mobility constraints limit ability to queue for relief supplies • Increased risk of neglect during prolonged shortages
Floods				
• Increased caregiving burdens for sick or displaced family members • Loss of income from informal trade due to market inaccessibility	• Loss of fishing, transport, and casual labour opportunities • Psychological stress from inability to provide for families	• School closures disrupt learning and training • Loss of casual labour -increased idleness, crime, and drug use • Girls particularly vulnerable to early	• High susceptibility to waterborne diseases and cold exposure • Limited mobility restricts evacuation • Greater dependence on others for shelter,	• Severe mobility barriers during evacuation and accessing aid • High risk of injury and neglect in shelters • Limited access to emergency services

Women	Men	Youth	Elderly	PADs
• Contaminated wells force longer distances to find safe water • Elevated psychosocial stress	• Damage to tools and assets disrupts livelihoods	marriage and pregnancy	food, and medicine	
Pests and diseases				
• High exposure due to their role in managing subsistence farms, small livestock, and seaweed farming • Reduced food security and income from harvest losses • Limited access to pest-control technologies and extension services	• Income loss from crop and livestock declines • Increased stress due to reduced labour demand and food shortages • Higher pressure to diversify livelihoods	• Loss of agricultural casual labour opportunities • Increased unemployment and financial vulnerability • Greater exposure to negative coping behaviours	• Increased dependence for food and care during outbreaks • Reduced capacity to protect crops and livestock • Nutritional deficits worsen chronic illnesses	• High dependency on family for food and adaptive support • Limited access to adapted technologies and extension services • Increased vulnerability from household food shortages
Human wildlife conflicts				
• Restricted mobility due to fear of wildlife attacks • Loss of crops and livestock increases household food insecurity	• High risk of injury when guarding farms and livestock, often at night • Economic strain from repeated	• Increased risk when helping to guard farms • Reduced productivity and fewer livelihood opportunities	• Very limited capacity to respond to wildlife encounters • Increased dependence after	• Inability to flee or protect themselves during wildlife incidents • Extremely high dependence on caregivers for

Women	Men	Youth	Elderly	PADs
• Limited access to compensation processes	crop and livestock losses • Pressure to seek alternative income sources	• Higher vulnerability to migration and exploitation	livestock or crop losses • Heightened food insecurity	safety and recovery • High vulnerability during compensation or reporting processes
Environmental Degradation				
• Reduced access to firewood, mangroves, herbs, and NTFPs • Walk longer distances (5+ km) for resources leading to exhaustion • Less time for childcare and economic activities	• Declining fish stocks increase offshore travel distance and fuel costs • Higher fishing risks • Reduced forest resource availability increases livelihood strain	• Fewer forest-based livelihood options (beekeeping, nurseries, eco-tourism) • Increased involvement in illegal extraction when opportunities shrink • Rising idleness and economic vulnerability	• Mobility challenges increase dependency during resource scarcity • Physical strain from long distances to collect resources • Heightened vulnerability to malnutrition	• Limited access to degraded resource areas • Exclusion from resource-based livelihood opportunities • Increased reliance on caregivers
Sea level rise				
• Destruction of seaweed lines and racks reduces income • More time spent relocating production sites • Loss of financial independence	• Submerged landing sites and damaged gear reduce catch • Higher risk during offshore fishing • Increased household financial pressure	• Fewer opportunities in fishing, seaweed farming, and tourism • Increased economic migration	• Very limited ability to relocate • High health risks from flooding and contamination • Emotional distress from loss of ancestral land	• Extreme difficulty relocating to safer areas • Heightened exposure to contamination and displacement • Increased reliance on caregivers

Women	Men	Youth	Elderly	PADs
		• Higher stress and vulnerability during relocation		
Strong winds				
• Increased caregiving demand after home damage • Higher risk securing household assets during storms	• High danger at sea and gear damage • Increased risk of capsizing during strong winds	• High injury risk from collapsing structures or debris • Limited ability to seek safe shelter	• Difficulty accessing safe shelter quickly • High injury and exposure risk	• Severe challenges evacuating safely • High exposure to falling debris and structural collapse

4.1 Children's Vulnerability to Climate Hazards and Key Mitigation Measures

Children are among the most climate-vulnerable groups in Kwale County due to their rapid developmental needs and high level of dependency on caregivers and community systems for protection, nutrition, learning, and emotional support (UNICEF, 2022). Their smaller body size, underdeveloped immune systems, and limited capacity to independently anticipate or respond to environmental hazards significantly increase their susceptibility to climate-related shocks and stresses (WHO, 2020). Climate change disrupts the availability of essential resources such as clean water, nutritious food, secure shelter, accessible healthcare, and stable educational environments, foundational elements for child survival and well-being. As a result, even minor climate disturbances can translate into severe consequences for children's health, safety, and long-term development.

During droughts, children face heightened exposure to acute malnutrition, stunting, anaemia, and dehydration as household food stocks diminish and agricultural productivity declines. Water scarcity forces many families to rely on unsafe water sources, increasing the incidence of diarrhoeal diseases, typhoid, and other waterborne infections. In many communities, the responsibility of fetching water shifts to children, particularly girls, who walk longer distances under extreme temperatures, exposing them to fatigue, heat stress, and protection risks while simultaneously contributing to school absenteeism (UNICEF, 2022). Floods further compound these vulnerabilities by destroying homes, early childhood development centres, and schools, washing away learning materials, and damaging WASH infrastructure. Floodwaters heighten the risk of drowning, injuries, respiratory infections, and outbreaks of waterborne diseases such as cholera. Schools are frequently used as temporary shelters during extreme flooding events, disrupting learning cycles, interrupting school feeding programmes, and widening pre-existing educational gaps. Displacement following floods or storms exposes children to psychosocial trauma, family separation, and increased protection risks, particularly in overcrowded and poorly regulated temporary shelters (UNFPA, 2021).

In coastal areas, environmental degradation and coral bleaching significantly undermine child nutrition by reducing the availability of fish, a critical protein source for many households. The loss of mangroves and coastal forests also removes natural buffers that protect communities from strong winds, storm surges, and flooding, increasing the physical exposure of children to these hazards. Human–wildlife conflict, particularly in areas adjacent to protected lands, presents additional risks. Children may encounter wildlife while accompanying caregivers to fetch firewood, guard farms, or graze livestock, exposing them to both physical harm and trauma (UNEP, 2022). The impacts of sea level rise and coastal erosion further heighten childhood vulnerability by forcing families to relocate repeatedly, undermining children's sense of safety, disrupting social networks, and creating emotional and behavioural challenges. Displaced children often experience anxiety, nightmares, loss of routine, diminished concentration, and regression in developmental milestones (UNICEF, 2021).

Across all climate hazards, poverty intersects with gender and age to deepen vulnerability. When climate shocks reduce household income, families may resort to harmful coping mechanisms that directly compromise children's rights and well-being. These include early marriage for girls, withdrawal of children from school to support domestic work or caregiving, child labour, and, in extreme cases, transactional survival behaviours among adolescents (UNFPA, 2022). Such strategies not only violate child protection standards but also entrench intergenerational cycles of poverty and climate vulnerability.

Protecting children from climate hazards therefore requires a comprehensive, multi-sectoral, and child-centred adaptation approach. Strengthening child-sensitive early warning systems is essential to ensure that households, schools, and community structures receive timely and

actionable information. Equally important is the enhancement of nutrition surveillance systems and the implementation of early action protocols to detect and treat malnutrition during droughts, food shortages, or livelihood disruptions. Climate-resilient WASH infrastructure, such as elevated latrines, borehole protection, and rainwater harvesting systems, is critical to preventing disease outbreaks during floods and prolonged dry periods. Ensuring that education infrastructure is safe, accessible, and climate-proof, including storm-resistant roofing, elevated classroom floors, and shaded play areas, helps maintain learning continuity during emergencies (UNESCO, 2021).

Equally vital is the provision of psychosocial support and trauma-informed child protection services during displacement and post-disaster recovery. Establishing child-friendly spaces, counselling services, and family tracing mechanisms reduces the emotional burden of climate shocks on children. Expanding social protection programmes, including child grants and school feeding initiatives, helps households avoid harmful coping strategies and stabilizes children's access to basic services during climate-induced economic stress (UNDP, 2022). Strengthening community-based child protection structures, including the capacity of teachers, community health volunteers, local leaders, and caregivers, ensures that children's safety and welfare are protected before, during, and after climate hazards.

5 Adaptation Action Plan

5.1 Strategic Adaptation Objectives and Thematic Priorities

To effectively reduce climate risks and enhance resilience in Kwale, the CCAAP adopts a set of strategic adaptation objectives organized around key thematic priority areas. These priorities align with the major hazards and vulnerabilities identified, and with national and county strategy frameworks (National Adaptation Plan (NAP), National Climate Change Action Plan (NCCAP) and County Integrated Development Plan (CIDP). Each thematic area encompasses specific goals and actions, described below, with an emphasis on sustainability, community ownership, and co-benefits (such as livelihoods and ecosystem health). Cross-cutting issues like capacity building, gender and social inclusion, and climate information services are mainstreamed throughout all themes.

Water Security and Resource Management: Ensure reliable and adequate water supply for domestic, agricultural, and ecosystem use in the face of drought and erratic rainfall. Key measures include developing new water sources (e.g. drilling and equipping solar-powered community boreholes in water-scarce areas), enhancing storage (construction of small dams, sand dams, and communal water pans to capture runoff), and promoting rainwater harvesting at household and institutional levels (installation of gutters and storage tanks). These interventions will be coupled with community-based water resource management to maintain infrastructure and equitably allocate water. By improving year-round water availability, the plan not only reduces drought impacts but also yields co-benefits in health (clean water access) and gender equity, women and girls benefit significantly from shorter water-fetching distances and times. Efficient water use will be promoted through irrigation technologies (drip irrigation, treadle pumps) to support agriculture even in dry seasons. Overall, strengthening water security is foundational to adaptation in Kwale, touching on livelihoods, health, and conflict reduction.

Climate-Smart Agriculture and Food Security: Protect and enhance agricultural production under a changing climate. This priority focuses on climate-smart farming practices to cope with droughts, floods, and pest pressures. Actions include dissemination of drought-tolerant and early-maturing crop varieties (e.g. improved cassava, sorghum, millet) and promotion of diversified crops. Farmers will receive training in integrated pest management (IPM) and improved agronomic techniques, such as water conservation tillage, mulching, and agroforestry, to restore soil fertility and reduce crop losses to pests/diseases. Establishing farmer field schools and demonstration plots will facilitate practical learning on pest surveillance, safe pesticide use, biological control methods, and climate advisories for planting timing. For livestock keepers, the plan promotes resilient practices like fodder production and storage (creating fodder banks of drought-tolerant fodder grasses), introduction of hardy livestock breeds (e.g. Galla goats that tolerate heat and sparse pasture), and regular veterinary outreach and vaccination campaigns to prevent disease outbreaks in stressed conditions. These measures aim to stabilize yields and farm incomes despite climate stresses. By improving crop and livestock productivity, the county will bolster food security and reduce the risk of hunger during climate shocks. Additionally, surplus production (when good seasons occur) can be preserved, e.g. through community grain banks and improved storage (hermetic bags, silos), to serve as a buffer in bad years.

Resilient Infrastructure and Human Settlements: Climate-proof vital infrastructure and settlements against floods, storms, and other extremes. Given the heavy damage caused by flooding and extreme weather, the plan prioritizes upgrades to transport and drainage infrastructure to withstand these events. This includes constructing or raising critical bridges and culverts at flood-prone river crossings to maintain year-round connectivity, and improving road drainage systems (culverts, trenches) especially along rural access roads so that heavy rains do not wash out roads or isolate communities. Flood control structures will be built where needed

e.g. reinforced riverbanks with gabions or levees in flash flood hotspots, and desilting of clogged drainage channels and rivers to restore natural flow capacity. The county will also establish or strengthen community early warning systems (EWS) for floods and storms, integrating KMD weather alerts with local volunteer networks to disseminate warnings via SMS, radios, and sirens. Identified safe havens (schools or churches on higher ground) will serve as emergency shelters during floods. In coastal and wind-exposed areas, building codes will be updated and local artisans trained on wind-resistant construction techniques, for example, reinforcing roofs with hurricane straps, using better rafter anchoring, and constructing raised platforms for key buildings. By making homes, schools, water facilities, and roads more resilient, these actions reduce the direct damage from disasters and enable quicker recovery. They also protect development investments (so that gains in education, health, commerce are not undone by the next flood).

Ecosystem Restoration and Environmental Management: Protect and restore natural ecosystems as buffers against climate impacts and to sustain community livelihoods. Healthy ecosystems (forests, mangroves, rivers, coral reefs) are natural defenses that reduce climate risks – for instance, mangrove forests dissipate wave energy and storm surges, forests regulate water flow and rainfall, and vegetative cover prevents soil erosion and landslides. This priority involves rehabilitating degraded forests and watersheds (through tree planting and community-led afforestation on denuded lands) and conserving coastal ecosystems. A flagship action is mangrove restoration along the coastline and estuaries: the plan will support communities and partner NGOs to establish mangrove nurseries and replant mangroves in degraded areas (such as along Vanga). Sustainable use of mangroves (community woodlots for firewood and timber to relieve pressure on wild mangroves) will be promoted to ensure long-term protection. Similarly, riverbanks and wetlands will be restored with native vegetation to reduce flood runoff and maintain groundwater recharge. The plan also addresses wildlife and biodiversity management as part of ecosystem resilience, working with KWS and communities to preserve wildlife corridors and reduce human–wildlife conflict by maintaining natural habitats. For example, planting unpalatable buffer crops or hedges at farm-forest boundaries can discourage wildlife encroachment even as natural forage inside forests is improved. Marine ecosystems are included as well, supporting coral reef conservation and sustainable fishing practices via the local BMUs will help fish stocks and protect shorelines.

Diversified Livelihoods and Social Protection: Enhance community resilience by diversifying income sources and strengthening safety nets, with a focus on women's and youth empowerment. This priority recognizes that purely agricultural livelihoods are highly vulnerable to climate shocks, so households need alternative or supplementary income options that are more climate-resilient. The plan will promote livelihood diversification programs such as beekeeping, poultry rearing, aquaculture (fish ponds, seaweed farming), and climate-resilient cash crops (like cassava, sweet potato) especially for women's groups and youth enterprises. For coastal communities, support will be given to expand seaweed farming (training and providing improved farming gear and drying facilities) and to add value to fish catches (cooling, processing) to increase incomes. These activities can provide earnings even when rains fail or when fishing is off-season. Alongside this, the plan strengthens financial inclusion and risk transfer mechanisms: establishing Village Savings and Loan Associations (VSLAs), facilitating access to microfinance for adaptation initiatives, and rolling out weather-index insurance schemes for crops and livestock to help farmers recover quickly after a bad season. A social protection mechanism is envisaged, in partnership with national programs, to provide cash transfers or food-for-work in extreme drought periods (building on NDMA's existing drought contingency funds). Special attention is given to human–wildlife conflict mitigation under this theme, since it often impacts livelihoods. Measures such as wildlife barriers (solar-electric fences or live fences), community wildlife scouts training, and rapid compensation funds for wildlife

damage will be implemented in hotspot areas. Awareness campaigns will teach communities safer practices (e.g. avoiding crocodile areas, using chili fences to deter elephants). Integrating HWC mitigation into adaptation planning ensures that families (especially farmers near parks) do not lose their livelihood gains to wildlife impacts. Overall, by broadening income streams and cushioning against losses, this priority aims to reduce poverty and inequality, and ensure that vulnerable groups (women, youth, marginalized communities) have greater adaptive capacity and independence. Many of these initiatives will be led by community groups with support from NGOs and the county's enterprise programs, thereby also building local entrepreneurship and skills.

Gender mainstreaming and community governance: Ensure that climate adaptation is inclusive, equitable, and anchored in strong local institutions. It will integrate gender justice as a guiding principle to guarantee that women, men, youth, and Persons Abled Differently (PADs) participate meaningfully and benefit equitably from all adaptation actions. Recognizing that decision-making and resource control have historically been male-dominated, the plan will deliberately empower women to take active leadership roles in Ward Climate Change Planning Committees (WCCPCs), water resource user associations, and producer cooperatives, enabling them to influence local planning and resource allocation. The plan will further promote gender-responsive budgeting and build the capacity of women through leadership, advocacy, and entrepreneurship training to enhance their agency in adaptation governance. Community governance structures will also be strengthened to ensure accountability and sustainability, with the plan supporting local groups, such as Village Savings and Loan Associations (VSLAs), Community-Based Organizations (CBOs), and Beach Management Units (BMUs), to formalize their operations through registration, adoption of bylaws, and establishment of elected leadership and functional bank accounts to manage their finances transparently. By reinforcing these organized governance mechanisms, the plan will enhance community ownership, inclusivity, and financial integrity, ensuring that climate resources are equitably distributed and that women and other marginalized groups play a central role in driving locally led climate resilience across the ward.

Cross-cutting support: All the above thematic actions are supported by cross-cutting strategies in capacity building, climate information services, disaster preparedness, and inclusion. Public awareness and education campaigns will be carried out to inform citizens about climate risks and adaptation practices (e.g. water conservation). The plan will strengthen early warning systems for multiple hazards (drought, floods) and ensure timely communication to communities (through radios, SMS, community barazas). Continuous capacity building of local institutions (WCCPCs, CBOs) and county staff is planned so that they can effectively plan, implement, and monitor adaptation initiatives.

5.2 Adaptation Actions and Implementation Plan

The adaptation actions identified across Kwale County were largely uniform across the wards, reflecting similar climate risks and community priorities, except for the Construction and desilting of a community water pan along the Mutsongo–Kimori range, which was specific to Vanga Ward due to its unique topographical and hydrological conditions. Based on findings from the Participatory Climate Vulnerability Assessments (PCVAs), ward adaptation plans, and strategic priorities, a comprehensive framework of adaptation interventions to be implemented between 2025 and 2030 has been developed (Table 3). These actions respond to seven priority climate hazards; drought, floods, pests and diseases, strong winds, Environmental degradation, sea level rise and human–wildlife conflicts and are anchored in the Kenya Climate Change Act (2023), the National Adaptation Plan (2015–2030), the Kwale County Climate Change Act (2021), and national frameworks on gender equality, child protection, and sexual and reproductive health and rights (SRHR). Implementation follows a phased approach comprising

short-term interventions (≤1 year), medium-term measures (≤3 years), and long-term interventions (>3 years), all anchored in gender equity, social inclusion, SRHR continuity, and community ownership to ensure climate benefits are distributed fairly and sustainably.

Under Thematic Area One, Water Security and Management, the adaptation actions prioritise immediate, medium-term, and long-term measures that enhance safe water access, reduce the drudgery associated with water collection, and protect women, girls, youth and other vulnerable groups from exposure to human–wildlife conflicts at water points. Short-term actions emphasise household-level rainwater harvesting through the installation of gutters and storage tanks in villages across Pongwe-Kikoneni, Ramisi and Vanga Wards. These decentralised systems ensure that households, particularly those led by women or located far from communal sources, can store rainfall during wet periods and reduce reliance on unsafe or distant water points during droughts. Complementary actions include constructing a new majibao in Mkwiro to enhance community storage, installing an elevated storage tank at Dzibwage, and sensitising women and youth on safe water treatment practices to reduce household disease burdens.

Medium-term interventions expand institutional water security by installing gutters and high-capacity tanks in one school per village across the three wards. These systems are intended to reduce school absenteeism during dry periods, promote hygiene, and lighten the caregiving burden on women whose time is often diverted to water collection. Solar-powered water pumps are also introduced on existing wells in Pongwe-Kikoneni and Ramisi, with strategic solar pumping points proposed along Rivers Ramisi and Umba to reduce turnaround time and minimise women's exposure to human–wildlife conflicts.

Long-term actions advance durable water infrastructure, including drilling and solarising boreholes in key villages across the three wards and constructing water pans in Ramisi and Vanga to increase surface water storage. A piped water system extending from the mainland to Mkwiro–Wasini is proposed to address chronic water scarcity on the island, where scarcity disproportionately affects women and girls responsible for domestic water supply. Collectively, these actions aim to deliver a reliable, climate-resilient water system supported by county institutions, water service providers, NGOs and private sector partners.

Under Thematic Area Two, Food Security, Ecosystem Restoration and Livelihoods, the adaptation actions aim to strengthen local production systems, restore degraded ecosystems and diversify climate-resilient livelihood pathways in a manner that is gender-responsive and socially inclusive. Short-term measures focus on capacitating women and youth through training on Resilience Design techniques such as soil fertility enhancement, water retention structures like Zai pits, and labour-efficient farm layouts, to increase productivity with minimal external inputs. Complementary training on Climate Smart Agriculture equips farmers with skills on drought-tolerant crops and improved livestock feeding practices to withstand increasingly erratic rainfall. Early action also includes establishing tree nurseries, promoting agroforestry, sensitising farmers on crop and livestock insurance and initiating mangrove restoration in shoreline villages.

Medium-term actions build on these foundations by supporting lead farmers especially women and youth champions, to fully implement and demonstrate resilient design and CSA practices. These demonstration farms enhance peer learning and adoption across the wards. Extension services and provision of certified seed varieties further strengthen production systems, while fodder seed banks and the introduction of drought-tolerant livestock breeds support pastoral households. Regulatory actions such as sand-harvesting control along Rivers Ramisi and Umba aim to reduce riverbank degradation. For coastal ecosystems, technical support to mangrove

nurseries ensures quality seedling supply for ongoing restoration efforts, while improved seaweed farming technology and post-harvest handling strengthen marine-based livelihoods.

Long-term initiatives emphasize the establishment of Farmer Field Schools across the three wards, large-scale terrestrial and mangrove ecosystem rehabilitation, implementation of Participatory Forest Management Plans through CFAs, and expansion of value-chain opportunities in agriculture, livestock, forestry and coastal resources. Support to seaweed farmers through advanced farming systems and expanded market access further contributes to diversified, climate-resilient livelihoods. Together, these actions promote food security, ecological stability, and sustainable income generation across households.

Under Thematic Area Three, Diversified Livelihoods and Social Protection, the actions strengthen climate-resilient income options while cushioning vulnerable groups during climate shocks. Short-term efforts prioritize training women, youth and PADs in alternative livelihoods such as kitchen gardening, beekeeping, poultry farming and small-scale ecotourism to reduce dependency on climate-sensitive livelihoods such as seaweed farming. Enhanced training in swimming and snorkelling for seaweed farmers promotes safety and enables access to deeper, more productive farming areas.

Medium-term actions support the establishment of these alternative enterprises through the provision of start-up materials and equipment, including horticultural inputs, modern beehives, incubators, boats for ecotourism and safety gear for marine activities. Relocation of seaweed farms to deeper waters further stabilizes marine livelihoods, while governance strengthening of women and youth VSLAs enhances household financial buffers and adaptation capacity.

Long-term actions prioritise the development of nature-based income models, including support for blue carbon project registration in Ramisi and Pongwe-Kikoneni, aligned with national carbon market regulations. Construction of a community social hall in Vanga provides additional income opportunities and a centre for community learning and engagement. These actions collectively enhance household resilience and enable diversification that mitigates both climate and market risks.

Under Thematic Area Four, Community Infrastructure and Disaster Management, adaptation actions aim to reduce exposure to climate hazards, strengthen emergency preparedness and enhance the resilience of community infrastructure. Short-term actions include training households and artisans on resilient building practices to reduce wind damage, alongside tree planting to act as windbreaks in villages highly exposed to strong winds.

Medium-term interventions introduce food-for-work programmes during drought, ensuring youth engagement in public works such as drainage restoration, culvert cleaning, gabion construction and tree planting to reduce flood and erosion risks.

Long-term measures include constructing raised footbridges and culverts in flood-prone crossing points, improving road drainage, designating disaster assembly areas and equipping community emergency response teams with appropriate tools. Expanded early warning systems and improved sanitation infrastructure such as household pit latrines in Ngweneni further strengthen community preparedness and disease prevention during flood events.

Under Thematic Area Five, Human–Wildlife Conflict Resolution, the adaptation actions focus on reducing conflict risks and strengthening coexistence between communities and wildlife. Short-term measures prioritize sensitization on conflict hotspots, safe behaviour and reporting mechanisms in villages near forests, rivers and wildlife movement corridors.

Medium-term actions introduce nature-based deterrents such as chilli cultivation and apiculture along forest-farmland boundaries, which help reduce crop destruction while promoting alternative income opportunities.

Long-term actions include installing sectional solar-powered fencing in crocodile hotspot areas, particularly in Kiwambale, to protect residents accessing rivers for water, fishing or livelihood activities. These actions aim to reduce loss of life, minimize injuries and enhance safety, particularly for women and youth who frequently use rivers for domestic activities.

Under Thematic Area Six, Gender and Social Inclusion, the adaptation actions ensure that climate resilience programming fully integrates protection, empowerment and meaningful participation of women, youth and PADs. Short-term actions include designing GBV anticipatory actions, strengthening awareness of GBV response services, and conducting youth boot camps that address drugs, antisocial behaviour and reproductive health.

Medium-term actions enhance gender-responsive governance through capacity building and strengthening VSLAs as safe spaces for information exchange and financial empowerment. Procurement of GBV response mechanisms including survivor transport and clinical response kits ensures rapid support when climate-related stressors heighten GBV risks.

Long-term actions include establishing a GBV recovery centre, constructing a drug rehabilitation center for affected youth, and sustaining school feeding and scholarship programmes to reduce child vulnerability during climate shocks. PAD-focused support ensures equitable access to assistive devices, livelihood opportunities and climate adaptation benefits across all wards.

Table 3: Kwale County Adaptation Action and Implementation Plan

Adaptation actions				
Short-term	**Medium**	**Longterm**	**Potential implementing partners**	**Budget (KES)**
Thematic area 1: Water Security and Management				
Rainwater harvesting through Installation of gutters on house roofs and water tanks at household level in Pongwe-Kikoneni (Anzuwani, Kibuyuni, Kiwambale, Majoreni Barabarani and Mkwiro villages), Ramisi (Darigube, Dzibwage, Mafisini, Mwagundu and Nyumba Sita villages), and Vanga (Mikoroshoni, Mwalewa C, Ngweneni, and Sega D Villages) Wards	Rainwater harvesting installation of gutters and high-capacity water tanks (10m^3) in schools – one school in each of the village per Ward (Pongwe-Kikoneni 6, Ramisi 5 and Vanga 4) to enhance water access by women and children	Drilling and solarizing boreholes in Pongwe-Kikoneni (Kiwambale and Majoreni Barabarani villages), Ramisi (Darigube, Dzibwage, Mafisini, Mwagundu and Nyumba Sita villages) and Vanga (Mikoroshoni, Mwalewa C, Ngweneni and Sega D villages) Wards	Kwale County Government-FLLoCA, KWAWASCO, NGOs and Private Sector Possible partnership with Kenya Ports Authority (KPA) Shimoni Port on Mkwiro Water Pipeline	50,000/HH, 400,000/ school, 70,000/ water pump, 5,000,000/ Borehole, 3,000,000/ Water pan, BOQ for Majibao & water pipeline in Mkwiro
	Construction of a new majibao storage in Pongwe -Kikoneni Ward (at Mkwiro village) to reduce water scarcity and workload among women and girls	Construction of water pipeline and kiosks to supply Mkwiro – Wasini Island		
	Installation of solar powered water pumps on existing wells to reduce women workload (turn-around time) in Pongwe-Kikoneni (Anzuwani, Kibuyuni, Kiwambale, Majoreni and Barabarani villages) and Ramisi (Darigube, Dzibwage, Mafisini, Mwagundu and Nyumba Sita villages) Wards	Construction of a water pan in Ramisi (Darigube village) and Vanga (Mikoroshoni, Mwalewa C and Ngweneni villages) Wards		
Installation of an elevated storage water tank at Ramisi Ward (Dzibwage village community water point)	Strategic (where embankment for water intake is viable) installation of solar powered water pumps at Rivers Ramisi Umba to reduce women exposure to HWC as well as workload (turn-around time) in Ramisi and Vanga Wards respectively		Kwale County Government, KWAWASCO, NGOs and Private Sector	1,000,000/ Dzibwage water, 70,000/ water pump

Adaptation actions				
Short-term	**Medium**	**Longterm**	**Potential implementing partners**	**Budget (KES)**
Sensitizing women and youth on water treatment in Ramisi (Darigube, Dzibwage, Mafisini, Mwagundu and Nyumba Sita villages) and Vanga (Mikoroshoni, Mwalewa C, Ngweneni and Sega D villages) Wards			Kwale County Government, CHPs, NGOs and Private Sector	Public baraza for the three wards at 280,000
Thematic area 2: Food Security, Ecosystem Restoration and Livelihoods				
Training and supporting women and youth on Resilience Design techniques from homestead to watershed level (Soil fertility improvement, water conservation/retention (e.g. Zai Pits), farm design- to maximize on time and use of local inputs to reduce farming costs)	Supporting select women and youth lead farmers to fully implement resilient design techniques including climate smart agriculture to serve as demonstration farms for continuing learning by other farmers to enhance sustainability of the interventions	Establishing a Farmer Field School in Pongwe- Kikoneni, Ramisi and Vanga Wards	Kwale County Government, KARLO, NGOs and Private Sector	5,000 per day per trainee, Farmer Field School (FFFs) 10,000 per farmer for a 12-month cycle
Training women and youth on Climate Smart Agriculture (CSA) (drought tolerant crops such as katumani maize, cassava, sunflower, groundnuts, cashewnuts, sorghum), silage/hay making to address livestock feed challenges during drought in Pongwe- Kikoneni (Anzuwani, Kibuyuni, Kiwambale, Majoreni Barabarani and Mkwiro villages), Ramisi (Dzibwage and Mafisini villages) and Vanga (Mikoroshoni, Mwalewa C,	Provision of farm inputs (certified seed varieties for drought tolerant crops, organic and inorganic fertilizers, pesticides) and extension services in Pongwe- Kikoneni (Anzuwani, Kibuyuni, Kiwambale, Majoreni Barabarani and Mkwiro villages), Ramisi (Darigube, Dzibwage, Mafisini, Mwagundu and Nyumba Sita villages) and Vanga (Mikoroshoni, Mwalewa C, Ngweneni, and Sega D Villages) Wards to trained women and youth groups	Aggregating women and youth farmers' produce to enhance farm produce marketability and enhance value addition and agribusiness opportunities	Kwale County Government, KARLO, NGOs and Private Sector	5,000 per day per trainee Production costs: Maize 55,000 per acre and Cassava 40,000 per acre 10,000 for livestock farmers per year or crop cycle (Free if it is provided by the government)

Adaptation actions				
Short-term	**Medium**	**Longterm**	**Potential implementing partners**	**Budget (KES)**
Ngweneni, and Sega D villages) Wards				
Sensitize and facilitate access to information on crop and livestock insurance (index-based drought coverage) targeting women and youth groups involved in CSA in Pongwe-Kikoneni, Ramisi and Vanga Wards	Establish livestock fodder seed banks, pastures, and introduce drought-tolerant livestock breeds e.g., Gala goats/Dairy cows for women and youth groups in Pongwe - Kikoneni (Anzuwani, Kibuyuni, Kiwambale), Ramisi (Dzibwage and Mafisini villages) and Vanga (Mikoroshoni, Mwalewa C, Ngweneni) Wards	Support seaweed farmers with improved farming gear, drying racks, and capacity building on advanced farming techniques (bamboo rafts, tubular nets) Pongwe- Kikoneni (Mkwiro and Kibuyuni villages) and Ramisi (Nyumba Sita village) Wards	Kwale County Government, KARLO, NGOs and Private Sector	Public baraza for the three wards at 280,000, 55,000 per acre for pasture, 15,000 per Gala goat, 100,000 per dairy cow 50,000 per farm, bamboo rafts and tubular nets
Facilitate/train women and youth to establish tree nurseries of priority species (Annex 1) for on-farm tree production (Agroforestry, woodlots and boundary planting) in Pongwe-Kikoneni (Anzuwani, Kibuyuni, Kiwambale, Majoreni Barabarani and Mkwiro villages), Ramisi (Darigube, Dzibwage, Mafisini, Mwagundu and Nyumba Sita villages) and Vanga (Mikoroshoni Mwalewa C, Ngweneni and Sega D villages) Wards	Provision of priority tree and fruit seedlings at household level Ramisi (Darigube, Dzibwage, Mafisini, Mwagundu and Nyumba Sita villages) and Vanga (Mikoroshoni Mwalewa C, Ngweneni and Sega D Villages) Wards Rehabilitate degraded forest areas in Ramisi (Buda Forest) and Vanga (Gonja forest and Kaya Sega) Wards Support Gonja Forest and Ramisi CFA to implement PFMP		Kwale County Government, KFS, KEFRI, CFAs, NGOs, Private Sector	5,000 per day per trainee, 5,000 per HH (100 seedlings per HH), 150,000 per acre for terrestrial forest rehabilitation
Sensitization and training women and youth on mangrove restoration (nursery establishment, transplanting, tending and monitoring) in Pongwe- Kikoneni (Anzuani, Kibuyuni, Kiwambale and	Assessment and provision of technical support to the existing mangrove nurseries to ensure quality and sustainable seedlings supply for restoration interventions in Pongwe-Kikoneni and Ramisi Wards	Implement an integrated mangrove restoration programme in Ramisi (Nyumba Sita village) and Pongwe- Kikoneni (Anzuwani, Kibuyuni, Kiwambale and Majoreni Barabarani villages) Wards to enhance shoreline	Kwale County Government, KFS, KEFRI, CFAs, NGOs, Private Sector	5,000 per day per trainee, 75,000 per nursery, 450,000 per acre for mangroves restoration

Adaptation actions				
Short-term	**Medium**	**Longterm**	**Potential implementing partners**	**Budget (KES)**
Majoreni Barabarani villages) and Ramisi (Nyumba Sita village) Wards		protection and ecosystem resilience		
Training women, youth and PADs on making organic briquettes in Ramisi (Darigube, Dzibwage, Mafisini, Mwagundu and Nyumba Sita villages) and Vanga (Mikoroshoni Mwalewa C, Ngweneni and Sega D Villages) Wards	Provision of energy saving jikos and kiln in Ramisi (Darigube, Dzibwage, Mafisini, Mwagundu and Nyumba Sita villages) and Vanga (Mikoroshoni Mwalewa C, Ngweneni and Sega D Villages) Wards		Kwale County Government, NGOs, Private Sector	5,000 per day per trainee, 3,000 energy saving jiko & 7,000 per Kiln
Sensitization on riverbanks management and flood control and sand harvesting impacts along Rivers Ramisi and Umba in Ramisi and Vanga Wards respectively	Regulation of sand harvesting along Rivers Ramisi and Umba in Ramisi and Vanga Wards respectively	Construct gabions and check dams along River Ramisi, drainage channels, and eroded sections to reduce flood velocity and stabilize banks	NEMA, WRA, Kwale County Government, WRUAs, NGOs, Private Sector	Public baraza for the three wards at 280,000 BOQ required for Gabions and check dams construction
Thematic area 3: Diversified Livelihoods and Social Protection				
Training women, youth on alternative livelihoods such as kitchen gardening (horticultural crops), eco-tourism, beekeeping and poultry farming to address seaweed farming vulnerability, in Pongwe-Kikoneni (Kibuyuni and Mkwiro villages) and Ramisi (Nyumba Sita) as well as enhance income in Vanga (Mikoroshoni Mwalewa C, Ngweneni and Sega D villages) Wards	Provision of alternative livelihoods establishment materials/Equipment; kitchen gardening (seeds and organic fertilizers), eco-tourism boat modern beehives and eggs incubators in Pongwe- Kikoneni (Anzuwani, Kibuyuni, Kiwambale, Majoreni Barabarani and Mkwiro villages), Ramisi (Darigube, Dzibwage, Mafisini, Mwagundu and Nyumba Sita villages) and Vanga (Mikoroshoni, Mwalewa C, Ngweneni and Sega D Villages) Wards	Supporting women and youth to initiate and register a mangrove blue carbon project (Plan Vivo Systems and Standards) similar to neighboring Vanga blue carbon projects and consistent with the Climate Change (Carbon Markets) Regulations, 2024 in Ramisi and Pongwe-Kikoneni Wards	NEMA, KEFRI, KFS, Kwale County Government, CFAs, NGOs, Private Sector	5,000 per day per trainee (3,000,000-Boat, 7,000/Modern beehive, 100,000/egg incubator, 10,000/snorkeling equipment, 110,000/ application fees & design document, 1,000,000/ feasibility study for Blue Carbon Project

Adaptation actions				
Short-term	**Medium**	**Longterm**	**Potential implementing partners**	**Budget (KES)**
Training women and youth involved in seaweed farming on swimming and snorkelling in Pongwe-Kikoneni and Ramisi Wards	Relocate seaweed farms to deeper waters and provide boats and snorkeling equipment in Ramisi (Nyumba Sita village) and Pongwe- Kikoneni (Kibuyuni and Mkwiro villages) Wards			
Train VSLAs on Governance and alternative regenerative investment opportunities such as green bonds and money market funds to diversify interest income from surplus savings	Promote VSLAs excess savings investments in reputable interest earning schemes such as the green bond by Safaricom in Pongwe-Kikoneni, Ramisi and Vanga wards	Construction of a community social hall in Vanga Ward (Mwalewa C village) to generate income	Kwale County Government, NGOs, Private Sector	5,000 per day per trainee BOQ required for social hall construction 100,000 per VSLA
Thematic area 4: Community Infrastructure and disaster management				
Conduct community training on resilient building practices to strengthen structures (use of nails instead of strings on rooftops) in Pongwe-Kikoneni (Kibuyuni and Mkwiro villages and Ramisi (Mwagundu village) Wards to address strong winds risks Undertake tree planting to address strong winds in Pongwe Kikoneni (Kibuyuni and Mkwiro villages) Ramisi (Mwagundu village) and Vanga (Sega D village) Wards	Design and implement a food for work programme during drought (this has been done before in the county) targeting the youth to restore drainage channels, clean culverts, construct gabions and plant trees for erosion and flood control in Pongwe-Kikoneni, Ramisi and Vanga Wards	Construct raised footbridges/culverts at flood-prone crossing points in Pongwe - Kikoneni (Mwalewa C, Ngweneni and Sega D villages) and Vanga Wards (Majoreni Barabarani village) Improve road drainage and install culverts to channel floodwaters in Pongwe- Kikoneni (Majoreni Barabarani village), Ramisi (Dzibwage, Mafisini and Nyumba Sita villages) and Vanga (Mwalewa C village) Wards Designate flood disaster assembly areas and temporary camps for	Kwale County Government, KERRA, NGOs and Private Sector	5,000 per day per trainee 5,000 per HH (100 seedlings per HH) BOQs required based on numbers and reputable training institutions

Adaptation actions				
Short-term	**Medium**	**Longterm**	**Potential implementing partners**	**Budget (KES)**
		vulnerable communities in Pongwe-Kikoneni Ward		
Train selected community emergency response teams in Pongwe-Kikoneni Anzuwani and Majoreni Barabarani villages), Ramisi (Darigube, Dzibwage, Mafisini and Mwagundu and Nyumba Sita villages) and Vanga (Mikoroshoni, Mwalewa C, Ngweneni and Sega D villages) Wards	Develop and operationalize Early Warning Systems (EWS) for floods (forecasts via chiefs, barazas, media) in Pongwe-Kikoneni (Anzuwani and Majoreni Barabarani villages), Ramisi (Dzibwage, Mafisini and Mwagundu villages) and Vanga (Mikoroshoni, Mwalewa C, Ngweneni and Sega D villages) Wards	Equip community emergency response teams (motorized boat, PPEs, first aid kits) in Pongwe-Kikoneni, Ramisi and Vanga Wards Construction of pit latrines at household level in Pongwe-Kikoneni Ward (Ngweneni Village) to address disease outbreaks	KMD, NDMA Kwale County Government, Ministry of Interior and National Administration Kenya Red Cross, NGOs and Private Sector	5,000 per day per trainee 980,000 EWS - for the three wards Public baraza for the three wards at 280,000 BOQ required for equipping community emergency response teams 100,000/pit latrine
Thematic area 5: Conflicts resolution (Human Wildlife Conflicts)				
Sensitize communities in Ramisi (Kiwambale village) and Vanga (Mwalewa C village) on potential conflict hotspots, minimizing human-wildlife interactions and reporting to KWS (hotline) as well as compensation procedures	Promote chili cultivation (provide training and quality seeds) and apiculture (provide training and hives) along forest–farmland boundaries as a Nature- Based Solution (NBS) Ramisi (Dzibwage and Mafisini villages) Ward	Installing sectional solar powered fences in crocodile hotspot areas Pongwe-Kikoneni Ward (Kiwambale village)	KWS, Ministry of Interior and National Administration, Kwale County Government	5,000,000- Death, 3,000,000- Permanent Disability, 2,000,000- Any other injury, Crop/Property destruction- compensation based on market rates, 5,000 per day per trainee, 350 per chilli seeds packet, 7,000 for beehive, 1,000,000/ km for electric fence
Thematic area 6: Gender and social inclusion to enhance adaptation action				
Design targeted GBV anticipatory actions for drought	Develop and implement a Pongwe-Kikoneni, Ramisi and	Establish GBV recovery center (this could be at the county level)	National Government, Kwale County	1,000,000 for consultancy services

Adaptation actions				
Short-term	**Medium**	**Longterm**	**Potential implementing partners**	**Budget (KES)**
and floods climate hazards in Pongwe-Kikoneni (Anzuwani, Kibuyuni, Kiwambale, Majoreni-Barabarani and Mkwiro), Ramisi (Darigube, Dzibwage, Mafisini, Mwagundu and Nyumba Sita villages) and Vanga (Mikoroshoni, Mwalewa C, Ngweneni, and Sega D) Wards	Vanga Ward GBV anticipatory action plan in collaboration with relevant partners		Government and NGOs, National Police Service, NGOs, GVRC, COVAW, HAK	Construction of GBV recovery center subject to a detailed Bill of Quantities
Assess the existing early warning systems to produce and transmit reliable and timely information and forecasts for the implementation of GBV anticipatory actions ahead of climate hazards	Strengthen gender-responsive governance- inclusion of women, youth and PADs through capacity building in Pongwe-Kikoneni (Anzuwani, Kibuyuni, Kiwambale, Majoreni-Barabarani and Mkwiro), Ramisi (Darigube, Dzibwage, Mafisini, Mwagundu and Nyumba Sita villages) and Vanga (Mikoroshoni, Mwalewa C, Ngweneni, and Sega D) Wards		Kwale County Government and NGOs	5,000 per person per day for training
Use of existing women associations such as VSLAs to train women and girls on their rights and GBV response strategies, helplines and recovery centers for women and girls as well as the role of men in peaceful conflict resolution	Procure/provide GBV response mechanisms such as transfer vehicles to health facilities and specialized first aid and rape kits to treat and preserve evidence for prosecution respectively		Kwale County Government, Civil Society, NGOs, Private Sector	30,000 per VSLA BOQ for GBV response mechanisms
Create awareness on existing GBV response strategies, helplines and recovery centers for women and girls	Procure/provide GBV response mechanisms such as transfer vehicles to health facilities and specialized first aid and rape kits to		National Government Department of Gender, Kwale County Government, National	KES 10,000 for first aid kits in placed in local dispensaries

Adaptation actions				
Short-term	**Medium**	**Longterm**	**Potential implementing partners**	**Budget (KES)**
	treat and preserve evidence for prosecution respectively		Police Service, NGOs, Gender Violence Recovery Centre (GVRC), Coalition on Violence Against Women (COVAW) Healthcare Assistance Kenya (HAK)	Public baraza for the three wards at 280,000
Organize and hold youth boot camps on guidance and counselling focusing on drugs, antisocial behavior and reproductive health	Promote vaccination of children and youth against diseases such as cholera which could arise and spread on account of climate hazards Support school feeding programmes and scholarships	Construct drug rehabilitation centre targeting the youth (this can be done at the county level)	National Government, NACADA, Kwale County Government, NGOs, Private Sector	100,000- Boot camp, Vaccination at local medical facilities, BOQ required for drug rehabilitation centre School feeding and scholarships needs assessment
Facilitate PADs to link with organizations that can provide infrastructural assistance such as wheelchairs, specialized training and entrepreneur skills			Kwale County Government, NGOs, Private Sector	50,000 for meetings

This implementation matrix will guide the roll-out of adaptation projects in a coordinated manner. Each action will be incorporated into annual work plans of the relevant county departments and into Ward Adaptation Plans for community execution. The responsible entities listed will form implementation teams or sub-committees as needed (e.g. a Water Project Team for boreholes, an Agro-Pastoral Team for farm interventions) to manage day-to-day activities. The Ward Climate Change Planning Committees will monitor progress on the ground and report any issues or delays to the County Climate Change Unit through the quarterly reports.

Several actions (such as water infrastructure, bridges, and coastal defenses) require technical design and feasibility assessments upfront. The county will leverage engineering staff and where needed external experts to conduct these studies in 2025–2026 so that projects are well-designed and meet standards (including environmental impact assessments for bigger projects, overseen by NEMA). Procurement of contractors for works will follow public procurement laws, with a preference for labor-intensive approaches that generate local employment (e.g. engaging youth/women groups in nursery operations, tree planting, and routine maintenance). For community-level actions, funds channeled via the CCCF or grants will be managed through community structures with transparency and accountability measures (WCCPC oversight committees will track use of funds and assets).

5.3 Financing and resource mobilization

Implementing the plan will require a combination of county budget allocation, climate funds, national programs, and partner support. The Kwale County Government is expected to allocate a portion of its development budget each year to priority adaptation projects (e.g. borehole drilling in Water Dept budget, irrigation schemes in Agriculture Dept, emergency funds in Disaster unit). As mandated by the Climate Change Act 2023, all county sectors must allocate at least 2% of their budget to climate change activities, which could translate to roughly KES 2–3 million per year per ward for adaptation if secured. The County Climate Change Fund (CCCF) is a dedicated source that will finance many community-prioritized actions; the WCCPCs will prepare proposals to tap the CCCF regularly. Over five years, an estimated KES 10 million (aggregate) could be drawn from the CCCF for ward-level projects in Kwale if well-coordinated.

National government programs provide additional avenues: the plan will align projects to initiatives like the National Drought Emergency Fund (for water projects and drought mitigation), the Kenya Climate Smart Agriculture Project (KCSAP) (for promoting resilient crops and farming practices), and the Kenya Livestock Insurance Program (KLIP) (for livestock insurance). Where applicable, the county will seek funding from the National Climate Change Fund (NCCF) once operational and continue engaging with the National Treasury's Climate Finance Unit to access any performance-based grants under FLLoCA.

Development partners and NGOs currently active in Kwale, such as the World Bank, UNDP, USAID, WWF, CARE, Plan International, are expected to play a key role in co-financing and technical support. Some have ongoing programs (e.g. World Bank FLLoCA and DRMAP, WWF's coastal ecosystem projects, Plan International's community climate preparedness initiatives) that the county will align with CCAAP actions to secure synergies. For example, NGOs may fund capacity building, community awareness, or provide materials (rainwater tanks, seedlings) to complement county efforts. The private sector, including companies like Base Titanium (operating in Kwale) and local industries, will be encouraged through public-private partnerships or CSR initiatives to invest in adaptation, for instance by sponsoring community water projects or providing micro-insurance for farmers. The plan also encourages innovative financing, such as exploring carbon credits or payment for ecosystem services for mangrove restoration, which could bring additional revenue to community projects.

The implementation will follow a flexible, learning-by-doing approach: annual work plans will be reviewed and updated based on progress, new climate information, and community feedback. The County Climate Change Unit, working with WCCPCs, will produce an annual implementation report that details what was done, funds used, and results achieved for each action. This report will allow the County Adaptation Planning Committee (as established by the County Climate Change Act) to re-prioritize or add new actions if needed, ensuring the plan remains dynamic and responsive through 2030.

6 Monitoring, Evaluation, Reporting, and Learning (MERL) Framework

6.1 Introduction

Monitoring, Evaluation, and Learning (MEL) are integral components of the County Climate Change Adaptation Action Plan (CCAAP). They ensure that adaptation interventions contribute effectively toward building the county's climate resilience and achieving its long-term sustainability goals. The MEL framework provides mechanisms for tracking progress, evaluating outcomes and impacts, and facilitating adaptive management through structured learning. It reinforces accountability to the public, development partners, and implementing agencies while ensuring coherence with national and sub-national adaptation reporting obligations under the Climate Change Act (2023).

The framework adopts a participatory and inclusive approach, combining scientific monitoring tools with community-based monitoring systems to capture both quantitative indicators and qualitative experiences. Monitoring focuses on tracking inputs and outputs, evaluation assesses effectiveness and impact, while learning ensures continuous feedback and improvement. By embedding MEL within institutional structures such as the County Climate Change Unit (CCCU), County Adaptation Planning Committee (CAPC), and Ward Climate Change Planning Committees (WCCPCs), the county will institutionalize evidence-based decision-making and ensure long-term sustainability of adaptation measures. The approach aligns with national frameworks including the National Adaptation Plan (NAP), National Climate Change Action Plan (NCCAP), and the County Integrated Development Plan (CIDP).

6.2 The County Theory of Change

The County's Theory of Change (ToC) outlines the logical pathways through which adaptation inputs and activities translate into outputs, outcomes, and long-term resilience impacts. It articulates how county-level investments in climate-resilient infrastructure, capacity building, institutional strengthening, and sustainable livelihoods collectively reduce vulnerability to climate hazards.

- **Inputs/Activities:** Financial allocations from the County Climate Change Fund (CCCF), national transfers, development partner support, and technical assistance will finance activities such as construction of resilient water infrastructure, promotion of climate-smart agriculture, enforcement of environmental bylaws, and establishment of ecosystem restoration programs.

- **Outputs:** Tangible products from these activities include functional water systems and irrigation infrastructure, trained farmers and pastoralists adopting improved practices, rehabilitated forest and mangrove ecosystems, enhanced early warning dissemination, and operational local climate institutions across all sub-counties.

- **Outcomes:** The expected short- to medium-term effects include improved access to safe water during drought periods, stabilized agricultural yields, diversified income sources, strengthened institutional coordination, and enhanced preparedness and response to climate hazards across the county.

- **Impact (Goal):** The overarching goal is to enhance the adaptive capacity, resilience, and socio-economic well-being of communities in the county. This means reduced exposure and sensitivity to climate hazards, strengthened ecosystem services, improved food and water security, and sustained livelihoods despite climate variability.

The ToC assumes enabling conditions such as consistent policy support, effective inter-departmental collaboration, and active participation of marginalized groups including women,

youth, and Persons Abled Differently (PAD). Gender and social inclusion remain central: by empowering women and youth in decision-making and livelihood diversification, the county enhances equitable resilience-building and intergenerational adaptation capacity.

6.3 Performance Indicators for Adaptation

The County's MEL system is built around performance indicators structured along the ToC levels; inputs, outputs, outcomes, and impacts. These indicators are harmonized with Kenya's National Adaptation Plan (NAP) and National Climate Change Action Plan (NCCAP) indicators, ensuring consistency in national reporting while responding to county-specific adaptation priorities.

Input/Output Indicators:

1. WATER SECURITY AND MANAGEMENT

- Number of household rainwater harvesting systems installed (gutters + tanks).
- Number of schools equipped with high-capacity rainwater harvesting systems.
- Number of water pans constructed or desilted.
- Length/coverage of pipelines and kiosks constructed (e.g., Mkwiro–Wasini).
- Number of manual and solar-powered pumps installed on existing wells.
- Number of women, youth, and PADs using solarised/manual hand pumps.
- Number and percentage of women, youth, and PADs in water management committees.
- Number of village water storage structures constructed (Majibao, elevated tanks).
- Amount of budget allocated and utilised for water infrastructure.
- Number of boreholes drilled, equipped, and solarised.
- Number of boreholes fully operational and supplying water year-round.

2. **FOOD SECURITY, ECOSYSTEM RESTORATION AND LIVELIHOODS**

- Number of farmers trained on Resilience Design techniques.
- Number of farmers trained on CSA practices (land prep, planting, pest control, post-harvest).
- Acreage under drought-tolerant crops supported with certified seeds.
- Number of Farmer Field Schools established and functional.
- Number of demonstration farms established by women and youth lead farmers.
- Number of farmers receiving inputs (certified seeds, fertilizers, pesticides).
- Number of households supported with organic manure production and composting.
- Number of farmers adopting terracing, zai pits, swales, and other soil/water conservation measures.
- Number of livestock fodder seed banks and pasture plots established.
- Number of households receiving improved livestock breeds (Gala goats, dairy cows).
- Number of farmers sensitised on crop/livestock insurance.
- Number of households supported with grain storage equipment (hermetic bags, grain banks).
- Number of village pest and disease surveillance teams established and trained.
- Number of households receiving indigenous/fruit seedlings.
- Number of community seedling distribution events held.
- Number of on-farm tree planting sessions conducted.
- Number of mangrove nurseries supported and maintained.
- Number of acres of mangrove restored through enrichment planting.
- Number of community trainings on mangrove protection and sustainable harvesting.
- Number of NBS windbreaks (Neem, Casuarina) planted across all villages.

- Number of environmental conservation awareness sessions held.
- Number of households adopting sustainable charcoal alternatives or clean energy.
- Budget allocated and utilised for restoration activities.

3.DIVERSIFIED LIVELIHOODS AND SOCIAL PROTECTION

- Number of women and youth trained in alternative livelihoods (eco-tourism, beekeeping, poultry).
- Number of modern beehives, incubators, or eco-tourism boats distributed.
- Number of households supported with livelihood start-up kits.
- Number of community groups formed for value addition and market linkages.
- Number of training sessions on business development, value addition, and financial literacy.
- Number of social protection initiatives implemented (food distribution, cash transfer mobilizations).
- Number of school feeding programmes supported.
- Budget allocations for livelihood diversification implemented.

4. COMMUNITY INFRASTRUCTURE AND DISASTER MANAGEMENT

- Number of culverts reconstructed or improved with proper drainage.
- Number of flood-prone crossing points fitted with raised footbridges.
- Number of youth mobilised in food-for-work programmes for drainage, gabions, and erosion control.
- Number of villages with functional flood Early Warning Systems (EWS) linked to KMD.
- Number of community dissemination channels established (chief barazas, noticeboards, SMS).
- Number of community emergency response teams trained.
- Number of community emergency response teams equipped (boats, PPE, first-aid kits).
- Number of designated disaster assembly points established and labelled.
- Number of community trainings on resilient building practices conducted.
- Number of households implementing safer roofing techniques (replacing strings with nails).
- Budget allocated and utilised for infrastructure upgrades (roads, culverts, drainage).

5. CONFLICTS RESOLUTION (HUMAN WILDLIFE CONFLICTS)

- Number of community sensitization sessions on human–wildlife conflict hotspots.
- Number of households supported with HWC reporting mechanisms (hotline awareness).
- Number of kilometres of solar-powered fencing installed in hotspot areas (e.g., Kiwambale).
- Number of community trainings on safe behaviour in crocodile, baboon, and snake-risk zones.
- Number of reported HWC cases logged and submitted to KWS for follow-up.
- Funding allocated and utilised for HWC mitigation (fencing, awareness, surveillance).

6. GENDER AND SOCIAL INCLUSION TO ENHANCE ADAPTATION ACTIONS

- Number of gender-responsive governance trainings conducted.
- Number and percentage of women, youth, and PADs participating in ward committees.
- Number of VSLAs supported with GBV awareness and rights training.

- Number of GBV anticipatory action sessions conducted for drought, floods, and strong winds.
- Number of GBV response kits (first-aid kits, rape kits) distributed to health facilities.
- Number of early warning systems integrating GBV alert messaging.
- Number of PADs supported with assistive devices, rehabilitation linkages, or livelihood equipment.
- Number of youth bootcamps conducted on drugs, behaviour change, and reproductive health.
- Number of awareness campaigns on GBV helplines and safe reporting mechanisms.
- Number of meetings with PAD support organizations facilitated.

Outcome Indicators:

1. WATER SECURITY AND MANAGEMENT

- Reduction in average time spent collecting water (minutes/day), disaggregated by women, youth, and PADs.
- Increase in household water storage capacity (litres/HH).
- Percentage of water facilities functional year-round.
- Increase in number/percentage of women, youth, and PADs with independent access to water sources (boreholes, solarised pumps, kiosks).
- Increased reliability of water supply during drought periods (days/months of service continuity).

2. FOOD SECURITY, ECOSYSTEM RESTORATION AND LIVELIHOODS

- Stability of crop yields over 3–5 seasons among supported farmers.
- Improvement in soil health indicators, including:
- Increase in soil organic matter (%)
- Increase in soil moisture retention capacity
- Improvement in soil nutrient status (N, P, K)
- Improved aggregate stability and soil structure
- Percentage of farmers adopting integrated soil and water conservation (terracing, zai pits, swales).
- Increase in acreage under diversified and climate-resilient farm systems.
- Increase in household-level adoption of post-harvest management technologies (grain banks, hermetic bags).
- Percentage of women, youth, and PADs participating in CSA and Resilience Design training cycles.
- Increase in on-farm and community tree cover (%) over baseline.
- Increase in mangrove seedling survival rates and total acreage restored.
- Reduction in incidents of illegal harvesting, charcoal burning, and mangrove cutting.
- Increase in households adopting windbreak and shelterbelt systems.
- Increased community participation in environmental conservation and sustainable harvesting practices.

3.DIVERSIFIED LIVELIHOODS AND SOCIAL PROTECTION

- Percentage increase in household income from diversified livelihoods (eco-tourism, poultry, beekeeping, seaweed value addition).
- Increase in the number of households with multiple income streams across seasons.
- Percentage of women, youth, and PADs accessing new livelihood opportunities.
- Increased financial literacy and business skills among trained groups.

- Increase in households benefiting from school feeding, cash transfers, or food-for-work schemes.

4. COMMUNITY INFRASTRUCTURE AND DISASTER MANAGEMENT

- Reduction in households exposed to flood-prone or erosion hotspots.
- Percentage of villages receiving timely Early Warning System (EWS) alerts.
- Community response rate to EWS alerts (attendance at assembly points, evacuation compliance).
- Reduction in flood-related damages to dwellings and access roads.
- Level of preparedness and readiness of community emergency response teams (measured via drills).
- Increased adoption of wind-resilient building practices in households.

5. CONFLICTS RESOLUTION (HUMAN WILDLIFE CONFLICTS)

- Reduction in frequency of human–wildlife conflict incidents (crocodile attacks, baboon raids).
- Increase in the number of households protected by solar-powered fences.
- Increase in community capacity to report and respond to HWC events (via KWS hotline).
- Increased community compliance with safe-behaviour protocols in HWC hotspots.

6. GENDER AND SOCIAL INCLUSION TO ENHANCE ADAPTATION ACTIONS

- Increase in the number and proportion of women, youth, and PADs participating in ward-level governance and adaptation committees.
- Reduction in drudgery for women, youth, and PADs due to improved water access and resilient infrastructure.
- Percentage increase in women's and PADs' access to credit, extension services, and productive resources.
- Increase in women's ownership of productive assets (livestock, seaweed farms, equipment).
- Improvement in community knowledge and awareness of GBV risk mitigation during climate hazards.
- Increased utilisation of GBV response services (safe shelters, first-aid kits, referral pathways).

Impact (Goal) Indicators:

1. WATER SECURITY AND MANAGEMENT

- Percentage of the ward population with year-round access to safe and reliable water.
- Increase in average litres of water available per person per day across villages.
- Long-term reduction in water scarcity–related vulnerabilities for women, youth, and PADs.
- Sustained decrease in waterborne diseases due to improved water quality and availability.
- Community-wide resilience to prolonged drought through diversified water sources and storage systems.

2. FOOD SECURITY, ECOSYSTEM RESTORATION AND LIVELIHOODS

- Sustained increase in crop yields above baseline across 3–5 year periods.
- Long-term reduction in seasonal yield fluctuations linked to droughts, floods, and pest outbreaks.
- Measurable improvements in soil health at landscape level, including:
- Higher soil organic carbon (SOC)
- Increased vegetation cover (NDVI index)
- Reduced erosion and sediment loss
- Enhanced household food security across all seasons.
- Increased biodiversity within integrated farming systems (trees, fodder, pollinator presence).
- Increases in forest cover, on-farm trees, and mangrove extent compared to baseline.
- Restoration of degraded landscapes evidenced by improvements in NDVI and tree density.
- Reduction in illegal harvesting, mangrove cutting, and charcoal burning.
- Enhanced ecosystem services, including shoreline protection, soil stability, and microclimate regulation.
- Long-term improvement in habitat quality for biodiversity and fisheries productivity.

3.DIVERSIFIED LIVELIHOODS AND SOCIAL PROTECTION

- Sustained increases in household incomes from diversified livelihoods (beekeeping, eco-tourism, seaweed, poultry).
- Improved household resilience to income shocks from climate hazards.
- Decreased poverty levels due to stable and multi-stream livelihoods.
- Increased school attendance and nutritional outcomes for children due to social protection schemes.
- Greater financial stability among women, youth, and PAD households.

4. COMMUNITY INFRASTRUCTURE AND DISASTER MANAGEMENT

- Significant reduction in climate-related damage to homes, roads, and community facilities.
- Communities demonstrating effective disaster preparedness and response during floods and storms.
- Long-term reduction in displacement, injuries, and asset losses during climate hazards.
- Increased structural safety and durability of settlements in strong-wind and flood-prone areas.

5.CONFLICTS RESOLUTION (HUMAN WILDLIFE CONFLICTS)

- Long-term reduction in human injuries, livestock losses, and crop destruction from wildlife.
- Improved coexistence between communities and wildlife due to effective mitigation (fencing, reporting systems).
- Reduced economic losses from HWC-related disruptions.
- Increased safety and mobility for women, children, and PADs in high-risk zones.

6. GENDER AND SOCIAL INCLUSION TO ENHANCE ADAPTATION ACTIONS

- Percentage increase in women's, youth's, and PADs' participation in climate governance and local decision-making structures.
- Strengthened continuity of SRHR, maternal health, and GBV response services during climate hazards.
- Increased leadership roles held by women, youth, and PADs in adaptation implementation.
- Growth in the number of marginalized and vulnerable groups leading or influencing climate adaptation actions.
- Reduction in GBV risks during droughts, floods, and strong winds due to anticipatory action systems.

Baseline data will be established using county-level assessments, socio-economic surveys, and historical climate datasets. Data will be disaggregated by gender, livelihood zone, and administrative unit to assess differential impacts and ensure inclusive adaptation outcomes. The County Climate Change Unit (CCCU) will coordinate data collection and analysis through a centralized digital monitoring system linked to national reporting platforms.

6.4 Adaptation Reporting and Learning

Reporting: The county will adopt a tiered reporting structure integrating ward, sub-county, and county levels. WCCPCs will compile quarterly progress updates feeding into sub-county adaptation reports, which the CCCU will synthesize into a county-level Annual Climate Adaptation Progress Report. This report will highlight achievements against indicators, emerging challenges, expenditure tracking, and gender-disaggregated performance metrics. It will also inform the County Annual Development Plan (ADP) and the State of the Environment (SoE) reporting.

Learning and Reflective Review: Mid-term (Year 3) and end-term (Year 5) evaluations will be conducted to assess performance, effectiveness, and sustainability of interventions. Participatory learning workshops will bring together county departments, communities, development partners, and academia to review progress, identify lessons, and revise priorities. Tools such as outcome harvesting, community scorecards, and Most Significant Change (MSC) narratives will be applied to capture qualitative insights. Findings will guide evidence-based revisions of the CCAAP and inform the subsequent planning cycles.

Knowledge Sharing and feedback to Policy: The county will institutionalize knowledge management through a digital climate knowledge hub, documenting case studies, research findings, and best practices. Cross-county exchange visits and inter-ward learning forums will promote horizontal learning, while upward policy feedback loops will relay local adaptation evidence to national institutions such as the National Climate Change Directorate (NCCD) and National Drought Management Authority (NDMA). This ensures county-level insights inform national adaptation strategies, fulfilling obligations under Section 25 of the Climate Change Act (2023).

Sustainability of M&E: To ensure continuity, the county will designate a Climate M&E Focal Officer within the CCCU to oversee data management, capacity building, and integration of adaptation indicators into the County Integrated Monitoring and Evaluation System (CIMES). MEL responsibilities will also be mainstreamed into departmental workplans, ensuring adaptation tracking becomes part of routine service delivery.

7 Risk-Informed Development and Disaster Management Integration

Kwale's CCAAP takes a risk-informed development approach, meaning it integrates climate and disaster risk considerations into all development planning and ensures coordination with disaster management structures. This approach recognizes that adaptation and disaster risk reduction (DRR) are two sides of the same coin: long-term resilience-building must go hand in hand with preparedness for extreme events. Several mechanisms are put in place to achieve this integration:

Mainstreaming Disaster Risk Reduction into Adaptation Actions: Each adaptation intervention has been designed with disaster scenarios in mind. For instance, the water security actions (boreholes, dams) are not only for gradual climate change adaptation but also function as drought emergency buffers when rains fail. Similarly, strengthening houses and early warning systems for storms is part of adaptation and directly reduces disaster impacts. The plan aligns with the NAP 2015–2030 directive to embed early warning systems and disaster preparedness into development. By doing so, development gains (like new infrastructure or agricultural productivity improvements) are protected from being wiped out by the next flood or cyclone.

Unified Local Committees for Climate and Disaster Management: At the ward level, the institutional arrangement deliberately overlaps the adaptation committee (WCCPC) with the Ward Disaster Risk Management Committee. In many wards, rather than having separate groups, the WCCPC either serves as the disaster committee or has a significant membership overlap (with additional co-opted members like local Red Cross volunteers or disaster response teams). This ensures seamless transition between "normal times" (where the focus is on resilience projects) and "disaster times" (where focus shifts to emergency response). For example, the same local committee that implements flood control measures also spearheads evacuations or relief distribution during an actual flood. Clear protocols will define roles: when a drought is looming (as signaled by NDMA early warning bulletins), the WCCPC/disaster committee will activate contingency plans such as water trucking or food aid linked with the national safety net programs. The integration avoids duplication of structures and information gaps, the people planning adaptation are the same who manage immediate response, leading to more coherent risk management.

Climate Information and Early Warning Systems: A core component of the plan is enhancing early warning systems (EWS) for climate hazards as a bridge between adaptation and DRR. The county, through KMD and NDMA, will ensure that seasonal climate forecasts, drought early warning stage updates, and extreme weather alerts are rapidly communicated to communities in user-friendly formats. The plan supports training of local leaders and volunteers on how to interpret these advisories and take early actions (e.g. plant early-maturing crops if rains are predicted to delay, pre-position animal feed if a drought warning is issued, or evacuate to safe ground when heavy rainfall/flood alerts come). Every year before the rainy seasons, the WCCPCs and County Disaster Management team will hold pre-season planning meetings to review the weather outlook and agree on preparedness measures (cleaning drains, alerting farmers, updating stockpiles of relief items). This operationalizes the concept of anticipatory action, which is advocated in Kenya's DRM policy and the NAP. For sudden-onset events, the integration with disaster response means that as soon as, say, a flood hits, the adaptation structures can document losses and start planning build-back-better recovery (like rebuilding houses with improved designs).

Development Planning with Risk Lens: The County government will use climate and hazard risk data to inform where and how development projects are implemented. For example, infrastructure projects in CIDP (roads, schools, water systems) will undergo climate risk screening, if a proposed road goes through a floodplain, the design will include sufficient

drainage and elevation (or be rerouted if too high-risk); new water supply schemes will consider future rainfall variability in sizing storage. Settlements expansion plans near the coast will incorporate sea-level rise projections to avoid placing critical facilities in zones likely to be inundated by 2050. This risk-informed planning is supported by the vulnerability maps and data from the ward assessments. The County Department of Planning, in concert with the Climate Change Unit and NEMA, will ensure Environmental Impact Assessments (EIAs) for major projects explicitly address climate and disaster risks, as now required by regulations. Through these steps, climate-proofing development becomes a standard practice rather than an afterthought.

Linkages with Disaster Response Mechanisms: The CCAAP explicitly links with existing disaster response funds and protocols. The National Drought Emergency Fund (NDEF) and other rapid response funds (e.g. Kenyans for Kenya, Kenya Red Cross emergency funds) will be activated as per NDMA's drought phase classification to support actions like water trucking, food relief, or destocking during severe drought, these are written into the drought contingency plans at ward level. The plan's MERL will track not just adaptation indicators but also disaster loss data (people affected, economic losses from events) to gauge if adaptation efforts are reducing these over time. Furthermore, the adaptation projects are designed to have dual use: for instance, rainwater harvesting tanks installed as adaptation also serve as emergency water sources in a sudden water crisis; grain banks set up to preserve surplus can double as food distribution hubs in a famine, etc. The presence of trained community responders (through Red Cross and St. John Ambulance collaborations) means that when a disaster strikes, local capacity is there to act while awaiting external help. Overall, the plan fosters a culture of prevention and preparedness in line with the Sendai Framework for DRR, by building local systems that anticipate hazards and minimize impact, rather than purely reacting afterward.

Continuous Risk Assessment and Scenario Planning: The County Climate Change Unit, in partnership with NDMA and KMD, will annually update a Climate Risk Register for the county, essentially a living document that identifies emerging risks (e.g. new disease vectors, potential landslide zones due to land use changes, etc.) and flags any adjustment needed in the adaptation plan. They will conduct scenario exercises, for example, "what if the next short rains fail completely?" or "what if a cyclone makes landfall near Kwale?" and use those to stress-test the readiness measures. This kind of forward-looking approach ensures that the adaptation actions remain relevant under a range of future conditions (including extreme worst-case events). The learning from actual disaster events will also feed into adaptation: if a flood causes damage in a new area, that area will be targeted for future flood control investment.

The disaster risk management is interwoven through the CCAAP, creating a resilient development pathway for Kwale. By having joint committees, shared early warning systems, and risk considerations in development, the county will be able to protect development gains, reduce losses from climate extremes, and bounce back faster when events occur. This integrated approach is in full alignment with national priorities on Disaster Risk Management integration (as highlighted in NCCAP and the National Disaster Risk Management Policy) and positions Kwale as a model for local climate-resilient development planning.

8 References

1. Vanga Ward Adaptation Plan (2025) – Participatory Climate Risk and Vulnerability Assessment Findings and Adaptation Actions.
2. Ramisi Ward Adaptation Plan (2025) – Participatory Climate Risk and Vulnerability Assessment Findings and Adaptation Actions.
3. Pongwe Kikoneni Ward Adaptation Plan (2025) – Participatory Climate Risk and Vulnerability Assessment Findings and Adaptation Actions.
4. Kenya National Adaptation Plan (2015–2030) – National Climate Change Priorities and Mainstreaming Guidelines.
5. National Climate Change Action Plan (NCCAP) 2023–2027 – Priority Adaptation Actions and Financing Mechanisms (FLLoCA, etc.).
6. Kwale County Climate Change Action Plan (2023–2027) – County Climate Change Act 2022 provisions, County Climate Change Fund, and Sectoral Targets.
7. Kwale County Integrated Development Plan (CIDP) 2023–2027 – Development Programs Mainstreaming Climate Resilience and Disaster Risk Management. (Kwale County Planning Unit, 2022).
8. Participatory Climate Vulnerability Assessment Reports (2025) – Community hazard timelines, vulnerability scores, and adaptation proposals in Vanga, Ramisi, Pongwe.
9. County Climate Change Act (Kwale County, 2022) – Establishment of Climate Change Unit, Ward Committees, and Integration of Climate Change in County Functions.
10. Kenya Climate Change Act (2023) – Devolved Climate Finance and Reporting Requirements for Counties.

9 Acknowledgements

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